

REVIEW ARTICLE

Exploring the Factors That Promote a Balance Between Academic Integrity and the Effective Use of GenAI Tools in Higher Education: A Systematic Review

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ABSTRACT

Background: This study explored the factors that influence the balance between academic integrity and the effective use of GenAI tools in higher education. It focused on the role of institutional guidelines in enhancing the responsible use of GenAI technologies to enhance academic integrity.

Objectives: The study was theoretically grounded in the Technology Acceptance Model and the Theory of Planned Behaviour to investigate the factors that promote academic integrity in using GenAI tools (RQ1), their impact and institutional strategies to effectively mitigate ethical risks (RQ2) and the model practices to support the ethical and effective use in higher education (RQ3).

Methods: The PRISMA framework was used to systematically review and thematically synthesise the results of 213 peer-reviewed articles published between January 2021 and May 2025.

Results: Finding indicates that academic support, defined by structured training, technical scaffolding, and perceived usefulness, is critical to enabling ethical GenAI use. Additionally, student self-regulation, as influenced by behavioural control and goal setting, was associated with greater academic integrity in GenAI-mediated learning. Whereas institutional policies varied widely, those with transparent, adaptive and discipline-responsive governance frameworks more effectively mitigated academic misconduct. Indeed, the model practices included GenAI ethics committees, interactive GenAI literacy modules, and the developer-educator collaborations to promote algorithmic transparency.

Conclusions: A comprehensive systems-based approach that encompasses academic support, self-regulation and ethical guidelines is critical for the responsible use of GenAI tools in education. Hence, to preserve academic integrity while nurturing innovation, institutions should integrate GenAI ethics into curricular design, faculty development and cross-sectoral policy frameworks. Future research may expand into multilingual and longitudinal analyses to support equitable and sustainable GenAI integration across diverse educational settings.

1 | Introduction

Integrating Generative Artificial Intelligence (GenAI) into higher education has sparked discussions on its transformative

potential and the ethical dilemmas it presents. GenAI comprises artificial intelligence technologies capable of generating text, images and other content using deep learning algorithms trained on extensive datasets (Rokhayani et al. 2022). Notable

Summary

- What is already known about this topic:
 - Generative Artificial Intelligence (GenAI) tools are increasingly used in higher education.
 - Academic support includes resource availability, training and technical guidance.
 - Self-regulation involves adaptive learning and personalised feedback.
 - Institutional regulations cover data privacy, bias and fairness guidelines.
- What this paper adds:
 - The study synthesises findings from 213 studies published between 2021 and 2025.
 - It highlights the importance of academic support, self-regulation and institutional regulations in integrating GenAI tools.
 - It emphasises the perceived usefulness and ease of use of GenAI tools in improving learning outcomes and skill development.
- Implications for practice and/or policy:
 - Institutions should provide comprehensive training, technical guidance and culturally relevant tools to enhance student learning and skills development.
 - Adequate academic support and self-regulation are essential for maximising the benefits of GenAI tools in higher education.
 - Clear institutional guidelines are necessary to ensure ethical standards in academic skills development.
 - Practitioners should focus on promoting the perceived usefulness and ease of use of GenAI tools to improve learning outcomes and skill development.

models, such as GPT-4 for language processing and DALL-E for image generation, exemplify its growing applications in academic contexts. These advancements enable personalised learning, automated content creation and real-time adaptive feedback, fostering new pedagogical possibilities (Burke and Crompton 2024). However, adopting GenAI tools in higher education necessitates carefully examining academic integrity considerations, governance frameworks and institutional responses to ensure responsible and effective use in educational settings.

One of the primary concerns surrounding GenAI in academia is its potential impact on academic integrity. While AI-driven technologies can enhance cognitive skills and research practices, their misuse, particularly in plagiarism, automated content manipulation and biased algorithmic outputs, raises ethical dilemmas (Skarżyńska et al. 2025). Institutions must navigate the challenges of maintaining integrity while embracing AI-enhanced learning tools. Lee and Lim (2023) have identified key factors influencing the ethical integration of GenAI tools, including institutional policies, faculty perspectives, student attitudes and technological capabilities. It suggests that GenAI tools may unintentionally facilitate dishonest academic behaviours without well-defined ethical frameworks and educational policies.

Hence, the impact of GenAI adoption on academic integrity extends beyond plagiarism concerns and reshapes knowledge

production and assessment strategies in higher education (Rodrigues et al. 2025). Traditional assessment models reliant on written assignments and exams may require reconfiguration to account for AI-assisted learning processes (Burke and Crompton 2024). In response, educational institutions have implemented strategies to mitigate ethical risks, including AI-use disclosure policies, faculty training programmes and AI-detection mechanisms. These institutional efforts reflect a broader movement toward fostering ethical AI literacy and ensuring students engage with GenAI tools responsibly. Such strategies also emphasise balancing innovation with rigorous academic standards to maintain integrity in AI-augmented learning environments.

Moreover, evidence-based best practices for ethically integrating GenAI tools in teaching and learning emphasise transparency, accountability and pedagogical adaptability. Studies highlight the effectiveness of structured guidelines on AI use, promoting academic honesty while enabling constructive AI applications in higher education (Skarżyńska et al. 2025). Faculty-led initiatives, such as critical AI literacy courses and research-based AI ethics discussions, have the potential to prepare students for ethical GenAI engagement (Lee and Lim 2023). Additionally, integrating AI-awareness programmes within curricula may enable students to evaluate AI-generated content and understand its limitations critically. These approaches illustrate how educational institutions can harness AI's benefits and promote an ethical learning environment grounded in integrity.

Therefore, GenAI's role in higher education demands a nuanced approach that embraces technological innovation and safeguards academic integrity. Hence, this systematic literature review explored the empirical insights from 2021 to 2025 to synthesise the results that would enable educators and policymakers to develop frameworks that ensure academic integrity and effective use of GenAI tools in teaching and learning. Institutional strategies, including policy refinement, faculty development, and AI-awareness initiatives, may provide avenues for responsible GenAI adoption. As higher education evolves in response to GenAI advancements, ethical considerations must remain at the forefront, shaping policies and pedagogical practices that preserve integrity in leveraging AI's educational potential.

1.1 | GenAI Use Conceptual Framework

This study integrates the Technology Acceptance Model (TAM) and the Theory of Planned Behaviour (TPB) to systematically explore the ethical and effective use of GenAI tools in higher education. These two were complemented by the Self-Regulated Learning Theory (Zimmerman 2002), which provides a structured foundation for evaluating the factors, impact and model pedagogical practices in the use of GenAI tools to preserve academic integrity in higher education, as illustrated in Figure 1.

The TAM by Davis (1989, cited in Silva 2015) serves as the foundational theory for understanding the integration of GenAI tools in higher education. According to TAM, technology acceptance is influenced by Perceived Usefulness (PU), which is the extent to which GenAI tools are accepted to enhance the learning outcomes. The Perceived Ease of Use (PEOU) refers to the

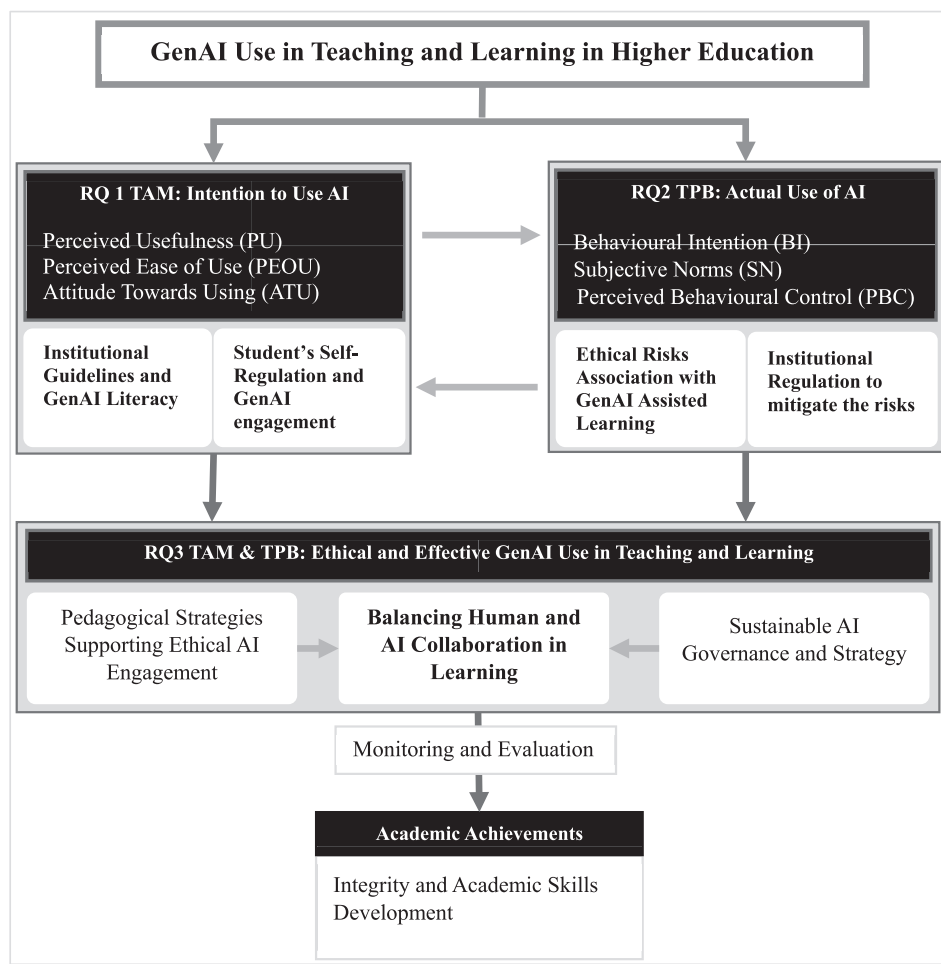


FIGURE 1 | Institutional and student self-regulated GenAI use conceptual framework.

accessibility and user-friendliness of GenAI tools for faculty and students. These TAM elements indicate a direct influence on the users' Attitude Toward Use (ATU), which is further shaped by institutional support, GenAI literacy initiatives and policy frameworks (Ajzen 1991). Hence, TAM predicts Behavioural Intention (BI) to adopt GenAI tools, which determines their actual usage patterns in educational settings. Thus, this framework was critical to investigate not only the technical adoption but also the proactive strategies that are embedded within academic curricula as the factors that promote the ethical use of GenAI tools.

Moreover, the TPB by Ajzen (1991) expanded the TAM, infusing the social and behavioural influences on technology usage. In the framework in Figure 1, Subjective Norms (SN), taken as the social and institutional pressures, are critical in shaping the attitudes toward the use of GenAI tools in education. In addition, the Perceived Behavioural Control (PBC) reflects the users' ability to regulate GenAI usage, influenced by university guidelines and ethical concerns. Hence, this study examined how academic institutions mitigate the ethical risks associated with GenAI-generated content, focusing on assessment redesign and integrity preservation. It was used as a lens to identify the institutional strategies that could mitigate the academic integrity risks and to explore which governance models were used to counter ethical challenges.

Furthermore, to ensure the responsible and impactful integration of GenAI technologies, the framework in Figure 1 synthesises insights from both TAM and TPB to establish the best practices for ethical GenAI use in education. The main considerations included the pedagogical strategies that foster GenAI literacy, the adaptive learning models that support the responsible use of GenAI tools, and the institutional policies that promote ethical reasoning and collaboration. Therefore, by integrating the TAM and TPB, the conceptual framework provides a basis for evaluating the empirical evidence on GenAI's role in higher education. Figure 1 outlines the three main elements that may help to identify the main factors that promote ethical GenAI use, to assess its impact on academic integrity, and to highlight the model practices for the responsible implementation in higher education.

2 | Literature Review

The integration of GenAI tools into higher education has generated extensive discourse regarding their impact on academic integrity. Indeed, as AI-driven platforms offer significant benefits in personalised learning, assessment automation and knowledge generation, they also introduce ethical concerns related to plagiarism, accountability and institutional monitoring. This review examines the literature reviews on the factors, impact,

and model practices in the integration of AI tools in higher education.

2.1 | Factors Promoting Ethical Integration of GenAI Tools in Higher Education

The empirical studies have emphasised the complexities of the ethical integration of GenAI tools in education by pointing to various digital literacy issues, institutional policy enforcement and the students' autonomy. According to Gavira Durón and Jiménez-Preciado (2025), significant AI literacy challenges are attributed to the inconsistent guidelines for ethical implementation across institutions. Additionally, in spite of educators and administrators acknowledging the expanding roles of GenAI tools in education, their response to addressing academic integrity issues remains fragmented, and they lack cohesive frameworks for integrating GenAI tools while upholding ethical standards.

Therefore, Bittle and El-Gayar (2025) examined the institutional strategies for mitigating academic dishonesty but predominantly focused on reactive measures, such as plagiarism detection tools, rather than proactive interventions like curriculum redesign or pedagogical adaptation. A more comprehensive approach would involve embedding GenAI ethics into faculty training and course design, ensuring that students engage critically with AI-generated content. Batista et al. (2024) synthesised the results of 37 studies published between 2023 and 2024 on AI-related assessment integrity and pedagogical strategies. However, their study did not include proposing concrete frameworks to mitigate emerging academic integrity risks. Similarly, Cai et al. (2024) highlighted concerns about educator preparedness and the absence of ethical safeguards in AI-mediated learning based on their synthesis of 158 studies published from 2021 to 2024. Indeed, their systematic literature review did not assess whether higher education institutions possess the requisite infrastructure to establish equitable GenAI governance mechanisms that could ensure responsible GenAI deployment in learning.

Moreover, Cai et al. (2024) introduced Vygotsky's Zone of Proximal Development as a theoretical framework for AI-assisted learning, suggesting that AI tools can scaffold students' intellectual growth, and highlighted concerns about educator preparedness and the absence of ethical precautionary measures. However, they did not sufficiently explore how faculty should regulate GenAI-driven assessments in higher education, which prompts the need for further discussion on ethical supervision in GenAI-mediated education, just as Rodrigues et al. (2025) argued that AI should serve as a complementary tool rather than a substitute for human instruction. Despite this, Rodrigues et al. (2025) also pointed to students' cognitive and behavioural challenges in distinguishing GenAI-generated content from original human work. Given the ability of GenAI tools to autonomously generate complex academic content, students' literacy in verifying GenAI outputs should, therefore, be more rigorously developed to ensure the integrity of adopting these tools.

Furthermore, Dimeli and Kostas (2025) emphasised the limited investigations into AI ethics within primary and

secondary education in their synthesis review of 50 articles published between 2023 and 2024. Although this highlights the need for interdisciplinary research on AI adoption across educational levels, it does not provide actionable recommendations for higher education institutions to develop institution-wide GenAI integration frameworks. Consistently, Pinto et al. (2023), based on their synthesis of 171 studies from 2018 to 2023, indicated the importance of institutional monitoring in machine learning adoption. However, they did not focus on the practical strategies for higher learning institutions to regulate GenAI-mediated education without compromising academic rigour.

2.2 | GenAI Adoption on Academic Integrity and Institutional Strategies for Mitigation

The integration of GenAI tools in higher education presents other challenges of originality, authorship, and integrity of academic practices, which necessitate structured institutional policies to mitigate academic quality risks. Naznin et al. (2025) explored AI writing tools, observing increased efficiency and concerns about the students' overdependence on these tools. Their overdependency may diminish their independent critical thinking and analytical reasoning skills. This issue reflects a wider challenge regarding GenAI's role in promoting cognitive engagement compared to enabling superficial automation of academic tasks. Newton and Jones (2024) evaluated traditional assessment methods, arguing that multiple-choice exams and essays increasingly become vulnerable to AI-generated responses that exceed human output. Their study advocates for critical thinking-based assessments on the one hand. However, their study does not propose systematic alternatives for ensuring authenticity in AI-assisted learning environments. Indeed, as GenAI output content continues to challenge traditional academic conventions, higher education institutions should reconsider whether existing assessment models can uphold academic integrity or call for curriculum reforms.

In addition, institutional responses to GenAI threats to academic integrity vary widely and are often reactionary rather than strategic. Sozon et al. (2024) outlined several intervention strategies, including implementing faculty training programmes, AI detection tools and educational policy updates. Nevertheless, their systematic review did not assess the long-term effectiveness of these measures. Similarly, Molina-Carmona and García-Peñalvo (2025) emphasised the importance of complementing AI's benefits with ethical governance, focusing primarily on institutional compliance rather than the student-centred approaches to responsible GenAI use. Moreover, Bittle and El-Gayar (2025) identified the academic challenges associated with GenAI-induced dishonesty and bias in their review of 41 studies published between 2021 and 2024. Still, they did not sufficiently explore institutional interventions designed to uphold academic integrity as they facilitate GenAI adoption in higher educational settings. Perkins and Roe (2024a, 2024b) examined 142 academic integrity policies from HEIs between 2022 and 2023, and they concluded that most institutions have yet to formulate explicit AI use guidelines in learning. Indeed, their study provides valuable

empirical insights into institutional unpreparedness but does not propose a structured model to address AI-mediated academic misconduct.

Furthermore, Imran and Almusharraf (2023) synthesised 30 studies from 2022 to 2023 to discuss challenges associated with AI-generated plagiarism. However, they did not provide a comprehensive evaluation of strategies that institutions can adopt to regulate AI-assisted writing. Equally, Singh and Ngai (2024), in their synthesis of 151 university responses from 2022 to 2023, highlighted varying institutional attitudes toward GenAI, in which they found that some perceive GenAI as a threat to academic integrity, while others advocated for its structured integration. Hence, the absence of consensus among HEIs underscores the need for the possibility of standardising integrity frameworks based on the synthesis of available empirical evidence as a solution that reconciles institutional strategy with AI innovation.

A recurring challenge in the literature is the assumption that AI-generated misconduct can be easily regulated through detection tools alone, overlooking the students' deeper cognitive and behavioural shifts associated with AI adoption in educational settings. Hence, effective mitigation strategies should extend beyond technical regulation and incorporate curriculum design adjustments that encourage students to engage critically with AI-generated content, promoting integrity, critical reasoning and accountability.

2.3 | Ethical Practices in GenAI Use in Teaching and Learning

The ethical and effective practices of GenAI integration in the teaching and learning process have been investigated by numerous empirical studies that advocate for responsible GenAI integration in education, but few offer concrete, evidence-based strategies beyond generalised ethical directives. For example, Chugh et al. (2025) discussed the role of GenAI tools in student learning engagement and proposed adaptive learning platforms that are designed to personalise learning pathways. However, they did not evaluate whether such platforms unintentionally promote passive learning rather than fostering intellectual rigour. Hence, GenAI tools' ability to automate content generation presents concerns about active student learning engagement when using GenAI tools to generate responses, warranting further investigation into strategies that should promote active learning and critical analysis.

Indeed, Allam et al. (2025) proposed the concept of living intelligence, which was explored to frame AI as a collaborative cognitive entity rather than a purely computational tool. This living intelligence concept aligns AI with the constructivist and sociocultural learning theories by Vygotsky (1978), advocating for the AI-human co-learning models. However, systematic review studies on its practical implementation remain scarce. Thus, further systematic empirical research is required to understand how the GenAI tools can facilitate meaningful student-instructor interactions while preserving pedagogical integrity. Additionally, Xia et al. (2024) proposed restructuring assessments to design them to cultivate self-regulated learning

practices in GenAI tool-integrated learning environments. However, their effectiveness in preserving academic integrity remains largely unexplored. Given AI's increasing roles in shaping educational experiences, HEIs must implement pilot studies to assess whether AI-driven assessments successfully enhance independent learning while safeguarding intellectual authenticity.

Moreover, Xia et al. (2024), based on a review of 32 studies, argued that HEIs lack transformative assessment models that promote integrity, self-regulated learning, and responsible AI use. Despite this, their study did not explain how GenAI-powered learning could complement traditional educational methodologies. Additionally, Kangwa et al. (2025) synthesised 148 studies from 2021 to 2024 and stressed the need for well-structured AI learning strategies. On the contrary, they did not examine the instructional models that effectively balance GenAI-enhanced learning with conventional pedagogy.

Furthermore, Melisa et al. (2025) reviewed 19 studies from 2023 to 2024, highlighting a need to understand GenAI's long-term impact on academic integrity regarding student motivation and critical reasoning skills. Though very insightful, the evaluation did not include how GenAI pedagogical and integration frameworks could mitigate GenAI's potential negative effects on students' independent and analytical thought. In addition, Ofusori and Hendradi (2025) synthesised 35 studies and emphasised the importance of institutional supervision in AI-assisted assessment without focusing on developing a scalable model for integrating AI tools into structured evaluation mechanisms. Therefore, it is still very important to:

- Identify the factors that promote integrity in integrating GenAI tools in higher education, focusing on institutional policies, educator preparedness and equitable AI governance models.
- Evaluate how the integration of GenAI tools affects academic integrity and examine institutional responses to mitigate ethical risks and preserve authenticity in student learning outcomes.
- Develop evidence-based model practices for GenAI-enhanced pedagogy that ensure that GenAI is responsibly integrated for better assessments, student learning engagement and knowledge retention.

Hence, adhering to the PRISMA framework, this study rigorously synthesised empirical findings to articulate the ethical and pedagogical strategies that provide higher education institutions with actionable recommendations for GenAI governance. The research questions that guided this systematic literature review included:

1. What factors influence the integrity of integrating GenAI tools in higher education based on empirical findings from 2021 to 2025?
2. How has the integration of GenAI tools impacted academic integrity, and what institutional strategies effectively mitigate ethical risks?
3. What evidence-based model practices support the ethical and effective use of GenAI tools in teaching and learning?

3 | Method

The study employed the systematic review methodology guided by the PRISMA (Page et al. 2021a) to explore three research questions. The research process involved searching, identifying and selecting articles following the PRISMA principles of extracting and managing secondary data (Lockwood et al. 2015). This systematic approach safeguards an unbiased and comprehensive data synthesis, aggregated and presented through synthesis tables and charts (Kangwa et al. 2024). Furthermore, the qualitative deductive and inductive coding methods were applied to analyse existing data and generate new themes (Aromataris and Munn 2024).

3.1 | Search Strategy

The designed search ensured the inclusion of high-quality and peer-reviewed studies to enhance the reliability and validity of the synthesised findings for this study (Page et al. 2021b). Hence, this systematic review focused exclusively on scholarly articles that were rigorously peer-reviewed and that the selected studies would contribute substantively to the discourse on GenAI integration in higher education. Additionally, the search criteria were restricted to the research articles published between January 1st, 2021, and May 31st, 2025. This publication period was selected because of the rapid evolution of GenAI technologies, necessitating an up-to-date review of the latest advancements. Additionally, the increasing integration of AI tools in higher education environments requires an examination of recent empirical and theoretical contributions to the integration processes. Thus, the temporal boundary targeted the most relevant findings and discussions that reflect technological developments and contemporary perspectives.

3.2 | Search Strings and Terms

The search strings and terms were systematically designed with a multi-layered approach to query formulation to enable an inclusive sample of relevant empirical studies. Article searches were conducted on PubMed, Web of Science, Scopus, ERIC and PsycINFO, as these electronic databases indexed rigorously peer-reviewed published research. The study followed the PRISMA principles to strengthen transparency, reproducibility and rigour in literature sampling (Page et al. 2021a). Indeed, snowball sampling techniques were additionally used to identify other publications, ensuring the inclusion of results that might not have emerged from the direct searches alone (Lockwood et al. 2015). Search queries were also thematically segmented to address database-specific constraints, enabling tailored strings across electronic databases with varying string limitations (Yusuf et al. 2024). For instance, Web of Science has a maximum of 50 search strings, which made the researchers to use a technique of structured decomposition of queries into distinct thematic categories based on the research question, such as 'AI ethics and integrity', 'institutional mitigation strategies' and 'evidence-based GenAI applications in teaching and learning', to optimise the retrieval efficiency.

Moreover, the search terms formulation integrated Boolean logic, using controlled term indexing and adaptive keyword

expansion to optimise precision and retrieval (Yusuf et al. 2024). The search strings were designed to accommodate generalised and tool-specific references to GenAI (Cai et al. 2024), thus capturing all the relevant studies. In addition to the core AI terminology, the search strings included prominent GenAI tools, such as ChatGPT and Gemini (Yusuf et al. 2024), to ensure the wider retrieval of empirical studies that explicitly discuss their adoption in academic integrity and pedagogical innovation. The search based on these search strategies yielded 4862 publications, and all of these underwent rigorous eligibility assessments to filter the results that are well aligned with this study's objectives (Page et al. 2021b). Table 1 presents the search strings and Boolean operators used for this study.

3.3 | Screening and Eligibility Criteria

The initial search result of 4782 articles across all the sampled electronic databases was subjected to the screening criteria that included publication dates from 2021 to 2025, peer-reviewed status, and removal of duplicates, and is presented in Table 2. The three researchers independently reviewed the titles and abstracts to determine eligibility, a process that resulted in a 100% interrater reliability, and a selection of 213 studies for further analysis. The initial discrepancies between the independent researchers were resolved through discussion and by adhering to the established eligibility criteria. The second stage in the screening process, after the screening on each electronic database, was the two researchers' manual screening of the articles by matching each record against the predetermined inclusion and exclusion criteria. Inter-rater reliability, calculated by percentage agreement, achieved a 95% agreement rate (Lewis et al. 2018). Any disagreements were again discussed until a 100% consensus was reached, and the articles that failed to meet quality standards were excluded, resulting in a final selection of studies that met the criteria for the systematic review, as in Table 2.

3.4 | Sample Extraction

The search results from electronic databases were retrieved through systematic file extraction to facilitate data organisation and citation management. Then the identified research articles were exported in multiple formats, such as CSV and RIS files, to optimise subsequent analysis and efficient bibliographic management through citation tools such as Zotero and Mendeley (Yusuf et al. 2024). Indeed, this approach streamlined the process of organising the sampled studies, sources and proper referencing for this study. The files were exported in CSV format from the electronic databases and merged into one Excel sheet. Then, data cleansing was done, and duplicates were removed. After removing duplicates, the eligibility of the remaining records was verified by looking through their titles and abstracts, which was performed independently by the researchers to prepare the eligible records for a possible full-text screening (Page et al. 2021a, 2021b). The differences in the independently selected studies were discussed and evaluated to reach a consensus among the researchers in line with the established eligibility criteria. This process ensured a 100% interrater reliability of the inclusion and exclusion procedure.

TABLE 1 | Search strings and Boolean operators.

Terms	Search strings
(a) GenAI	("generative artificial intelligence" OR "generative AI" OR "chatbot" OR "large language model" OR "LLM" OR "conversational agents" OR "GenAI" OR "ChatGPT" OR "Gemini" OR "AI-generated content" OR "transformer models" OR "GPT-3" OR "GPT-4" OR "language models") AND
(b) Education & Academic Integrity	("higher education" OR "universities" OR "tertiary education" OR "colleges" OR "university teaching") AND ("academic integrity" OR "plagiarism detection" OR "misconduct prevention" OR "ethical AI use" OR "faculty policies" OR "honour codes") AND
1. Institutional Mitigation Strategies	("Institutional Policies" OR "Guidelines" OR "Compliance" OR "Data Privacy" OR "Bias and Fairness" OR "Responsible Use" OR "Academic Integrity Oversight" OR "Curriculum Design" OR "AI Governance") AND
2. Self-regulation	("Self-regulation" OR "Goal Setting" OR "Progress Monitoring" OR "Academic Integrity" OR "Ethical Use" OR "Genuine Skill Development" OR "Behavioural Intention" OR "Accountability") AND
3. Evidence-Based Educational Models	("learning frameworks" OR "adaptive AI learning" OR "curriculum alignment" OR "faculty development" OR "ethical AI literacy" OR "academic support programs" OR "student engagement strategies") AND
(c) AI Skills Development	i. ("academic skills" OR "student development" OR "critical reasoning" OR "AI-enhanced learning" OR "student learning outcomes") OR ii. ("technical skills" OR "technical proficiency" OR "GenAI-assisted writing" OR "technical applications of AI") OR iii. ("ethical reasoning skills" OR "ethical AI use" OR "self-regulated learning" OR "academic accountability")

Note: Search was conducted as a combination of these strings in this format: (a) AND, (b) AND 'n' AND, (c) i. OR ii. OR iii. 'n' = string 1, OR string 2, OR string 3.

TABLE 2 | Inclusion and exclusion criteria.

Criterion	Inclusion	Exclusion
Language	Research Articles in English	Articles not published in English
Publication Type	Peer-reviewed full research articles	Review articles, books, book chapters, editorials, meeting abstracts, data papers, dissertations, preprints, and conference abstracts
Quality Score	High or medium-quality appraisal score	Low-quality appraisal score
Full-text Availability	Full text retrievable	Full text not available
Electronic Database	Indexed by PubMed, Web of Science, Scopus, ERIC, and PsycINFO	Not indexed by PubMed, Web of Science, Scopus, ERIC, and PsycINFO
Date of Publication	Published between January 1st, 2021 and May 31st, 2025	Published before January 1st, 2021 or after May 31st, 2025
Setting	Higher education	Other than higher education
Scope	Focus on GenAI, skill development, pedagogical strategies, and ethical implications	Not focused on these aspects

3.5 | Data Extraction

The data extraction process was adapted from Yusuf et al. (2024) and the PRISMA guidelines (Page et al. 2021a, 2021b). The two researchers independently and manually conducted the data extraction using a predefined coding scheme in Microsoft Excel, and then the third researcher validated

and refined the datasets. The sampled studies were systematically classified into distinct categories for in-depth thematic analysis. This data extraction process proceeded in three iterative phases, beginning with the initial coding, where titles and abstracts were reviewed for relevance; followed by the full-text analysis, which ensured comprehensive data representation beyond abstract summaries; and was finalised

by a cross-validation, to resolve the discrepancies through consensus-based review to strengthen the inter-coder reliability (Aromataris and Munn 2024; Lockwood et al. 2015). Indeed, the final dataset was consolidated into an Excel-based repository to enable qualitative thematic synthesis and quantitative trend analysis. Table 3 summarises the data extraction process to ensure a systematic alignment with research objectives while providing a strong foundation for data analysis.

3.6 | Quality Appraisal

The quality of the eligible studies was assessed using the Joanna Briggs Institute (JBI) checklist for Systematic Reviews and Research Syntheses (Page et al. 2021a, 2021b), which evaluates the trustworthiness, relevance, and results of published papers through 11 questions that assess various aspects of study design, conduct, and analysis (Moher et al. 2015). The main elements evaluated by the JBI checklist include proper randomisation of participants, allocation concealment, baseline similarity of treatment groups, blinding of participants and outcome assessors, and reliability of outcome measurements (Aromataris and Munn 2024; Lockwood et al. 2015). Studies were assigned a quality score based on how many JBI checklist items they met and were categorised into high, medium or low quality. Initial quality appraisal was independently conducted by two researchers who evaluated the abstracts for suitability based on inclusion and exclusion criteria and the JBI checklist, with disputes resolved through discussion

with a third researcher until consensus was reached, achieving 100% inter-rater reliability. The appraisal process resulted in high-quality ($n=183$) and medium-quality ($n=30$) with no low-quality articles being included for analysis (Kangwa et al. 2024).

3.7 | Data Synthesis

The study adopted a thorough data synthesis approach, employing thematic and narrative synthesis methods in a meticulously structured two-stage process (Aromataris and Munn 2024). Initially, the data synthesis involved becoming acquainted with the data, generating initial codes, and then searching for, reviewing, defining, and naming the themes, culminating in producing a detailed report (Aromataris and Munn 2024; Lockwood et al. 2015). Following this, narrative synthesis was utilised to develop a preliminary synthesis by examining relationships within and between studies, evaluating the synthesis, presenting the results, discussing the findings in the context of existing literature and theoretical frameworks, and concluding with recommendations (Aromataris and Munn 2024; Lockwood et al. 2015).

The thematic coding process was conducted independently by two researchers who initially coded the data, identifying recurring themes about the objectives of this study. The themes were then synthesised to harmonise them into categories for a comprehensive analysis of the empirical findings. The third researcher then independently verified this process for their suitability and accuracy (Kangwa et al. 2024), as outlined in Table 4 below.

TABLE 3 | Data extraction categories and coding framework.

Category	Coding elements
Study Characteristics	Author(s), year, country/region, publication type, research design, sample size, participant demographics
Factors Influencing GenAI Integration Integrity	Risks to academic integrity, institutional policies, governance frameworks, technological constraints, and ethical implications of GenAI use in education
Impact on Academic Integrity and Mitigation Strategies	Challenges in academic integrity (e.g., plagiarism detection, authorship concerns), institutional interventions, pedagogical adaptation strategies, and compliance frameworks
Evidence-Based Practices for Ethical GenAI Use	Evaluated case studies, faculty and student perceptions, AI-enhanced learning environments, and benchmarks for responsible AI implementation

3.8 | Results Synthesis Procedure

The results synthesis process followed a two-stage strategy, integrating thematic and narrative synthesis to achieve a more comprehensive analysis (Aromataris and Munn 2024; Lockwood et al. 2015). A thematic synthesis was conducted during the first stage to familiarise the data, generate initial codes, and organise them into themes based on recurring patterns (Braun and Clarke 2006). The generated themes were then refined through an iterative review process to ensure they aligned with the research questions before creating the final thematic report (Suprpto et al. 2024). Moreover, the second stage involved a narrative synthesis of the preliminary findings being contextualised by exploring relationships within and across studies, assessing the strength of the synthesised results, and presenting them coherently (Braun and Clarke 2006). Additionally, this phase included a detailed discussion of the findings to draw conclusions and make recommendations based on the main insights and implications of the results, as summarised in Table 5.

3.9 | Data Filtering and Screening Results

The sampling procedure illustrated in Figure 2 indicates the systematic sample selection, which initially yielded 4782 studies. These were screened according to the predefined inclusion and exclusion criteria before exporting them from the electronic databases, of which 4480 studies were excluded, leaving 302

TABLE 4 | Initial thematic coding.

Research question	Initial themes
RQ1: What factors influence the integrity of integrating GenAI tools in higher education based on empirical findings from 2021 to 2025?	Institutional Integrity Technological Constraints Academic Adaptation
RQ2: How has the integration of GenAI tools impacted academic integrity, and what institutional strategies effectively mitigate ethical risks?	Academic Integrity Institutional Safeguards Pedagogical Strategies
RQ3: What evidence-based model practices support the ethical and effective use of GenAI tools in teaching and learning?	Best Practices in AI Integration Faculty and Student Perspectives AI in Learning Environments

articles that were exported for further eligibility assessment. At the end of the screening and quality appraisal processes, 213 studies were included in this systematic analysis, as summarised in Figure 2 below and in Table A4.

4 | Results

4.1 | Characteristics of Sampled Studies by Country of Affiliation

The characteristics of the sampled studies encompassed a diverse and broad geographical scope, of which the sampled articles represent affiliated contributions from 59 countries. The highest number of studies was affiliated with the United States of America ($n = 35$), Australia ($n = 23$), and the United Kingdom ($n = 21$). These were followed by a notable contribution from Canada ($n = 8$), China ($n = 7$), and India ($n = 6$). Moreover, several other countries, such as Indonesia, Pakistan, South Africa, Spain, and Vietnam, were represented by a moderate number of articles ($n = 5$ each), as illustrated in Figure 3.

Additionally, other countries, including Malaysia, Oman, and Saudi Arabia, each contributed 4 articles, respectively, Bangladesh, Hong Kong, and Iraq ($n = 3$ each), while Botswana, Bulgaria, Ghana, and others had a representation of 1 study each. This sample distribution may have been influenced by research capacity, access to funding, and institutional emphasis on academic integrity in integrating GenAI into higher education.

4.2 | Characteristics of Sampled Studies by Publication Year

The included articles, as shown in Figure 4, reveal a dynamic trend in research output on academic integrity in integrating GenAI tools in higher education. There has been a rapid increase from 2021 ($n = 1$) to a substantial increase in 2023 ($n = 28$) and a peak in 2024 ($n = 111$), signifying a heightened focus that has continued to rapidly increase in the first half of 2025 ($n = 70$).

4.3 | Synthesised Results by Research Questions

This study used a narrative synthesis to analyse the data and present the findings in this section, drawing on the most critical findings from the included studies to answer the three research questions. This methodology facilitated a comparative account of the results, arranged according to the PRISMA criteria on a thematic level (Aromataris and Munn 2024; Lockwood et al. 2015). The results are presented in text, figures and tables according to the order of the research questions.

4.3.1 | Factors Influencing the Ethical Integration of GenAI Tools in Higher Education

The ethical integration of GenAI tools in higher education is shaped by institutional governance (RQ1 synthesised in Table A1), faculty preparedness, student self-regulation and technological infrastructure. This synthesis critically evaluates empirical findings from 131 studies, detailing their thematic contributions and institutional implications for responsible AI adoption as in Figure 5.

4.3.1.1 | Institutional Governance and GenAI Policies. The institutional guidelines in higher education are a foundation for the assurance of ethical integration of GenAI tools, as they influence faculty compliance, student engagement and policy development. This was indicated by 35 of the sampled studies, that universities with structured GenAI governance models experienced higher faculty engagement, improved monitoring of GenAI-assisted learning, and had fewer incidents of academic misconduct (Alqahtani and Wafula 2025; Jin et al. 2025; Zlotnikova et al. 2025). Additionally, Sullivan et al. (2023) and Wilson (2025) found that institutions with clear ethical policies for GenAI integration reported higher transparency and reduced ambiguity on students' use of GenAI tools in coursework. However, disparities between GenAI regulations across institutions limit consistency in ethical AI adoption, creating challenges in cross-institutional learning collaborations as noted by Plata et al. (2023) and Verhoef et al. (2022).

On the other hand, Henry and Oliver (2022) highlighted that the institutions that lacked well-defined GenAI governance

TABLE 5 | Two-stage synthesis process.

RQ	Themes	Predefined codes	Stage 1: Thematic synthesis	Stage 2: Narrative synthesis
RQ1:	Institutional Integrity	Governance Models, Ethical Guidelines, Accountability Mechanisms	• Familiarising with the data • Generating initial codes • Searching for themes	• Developing a preliminary synthesis • Exploring relationships within and between the studies
	Technological Constraints	Algorithm Bias, System Limitations, Reliability of GenAI Tools	• Reviewing themes • Defining and naming themes • Producing the report	• Assessing the robustness of the synthesis • Presenting the results • Discussion of the results • Conclusions and recommendations
	Academic Adaptation	Faculty Training, Student GenAI Literacy, Curriculum Alignment		
	Academic Integrity	Plagiarism Detection, Ethical GenAI Use, Authorship Concerns		
RQ2:	Institutional Safeguards	Policy Development, Compliance Monitoring, Educational Frameworks		
	Pedagogical Strategies	GenAI-Assisted Learning, Critical Reasoning Development, Adaptive Assessments		
	Best Practices in GenAI Integration	Case Studies, Benchmarking, Interdisciplinary Collaboration		
	Faculty & Student Perspectives	AI Perception, Engagement, Ethical Adoption		
RQ3:	AI in Learning Environments	Personalisation, Adaptive Feedback, Human-AI Interaction		

structures faced uncertainty in policy enforcement and led to inconsistent faculty interpretations of ethical GenAI use. Indeed, Currie et al. (2023) and Rudolph et al. (2024) indicated that the absence of standardised guidelines complicated faculty decision-making when regulating GenAI-assisted assessments and increased the risks of undetected GenAI-generated plagiarism. Moreover, Bannister and Carver (2025) and Nikolic et al. (2025) found that some universities that had no explicit policies regarding GenAI tools tended to rely on outdated plagiarism detection strategies, which may inadequately address GenAI-assisted academic misconduct.

Therefore, higher learning institutions should standardise GenAI ethics policies to address these governance challenges and to

ensure that faculty and students receive clear regulatory frameworks that balance innovation with integrity. Indeed, establishing cross-institutional GenAI governance collaborations may be essential in developing the unified ethical guidelines that can support the faculty to navigate the GenAI-enhanced pedagogy whilst promoting ethical student engagement (Johnson et al. 2024; Rana 2025). Hence, developing the interdisciplinary GenAI policy committees comprised of faculty, administrators, and GenAI ethics specialists can ensure comprehensive governance models that align with academic integrity standards and emerging GenAI advancements.

4.3.1.2 | Faculty Readiness and GenAI Literacy. The sampled studies ($n = 32$) highlighted that faculty preparedness is another critical aspect to ensuring ethical GenAI-assisted pedagogy; however, the faculty members were often found to have insufficient GenAI literacy training, which implies uncertainties in the use of GenAI tools for instructional purposes. Findings by Dhamija and Dhamija (2025), Lund et al. (2025), and Symeou et al. (2025) indicated that faculty members expressed concerns about GenAI misuse, particularly in student assignments and assessments. Despite recognising the GenAI tools' transformative potential, many educators remained uncertain about the effective instructional strategies that would be used to integrate the GenAI tools without compromising intellectual rigour and academic originality (Alexander et al. 2023; Williams and Fadda 2023). On the other hand, Kostka and Toncelli (2023) and Malik and Terzidis (2025) reported that the faculty-led GenAI ethics workshops enhanced preparedness, offered practical guidance on ethical GenAI integration, plagiarism detection, and GenAI-assisted grading protocols.

Although GenAI training initiatives have been introduced in some institutions, Slimi and Villarejo-Carballido (2024) underscore that the inconsistencies in faculty participation and the limitations in policy implementation were widespread, which may have reduced the efficacy of these programs. Hughes et al. (2025) and McIntire et al. (2024) found that there is a need for mandatory GenAI ethics training that equips faculty members with standardised guidelines for GenAI-assisted

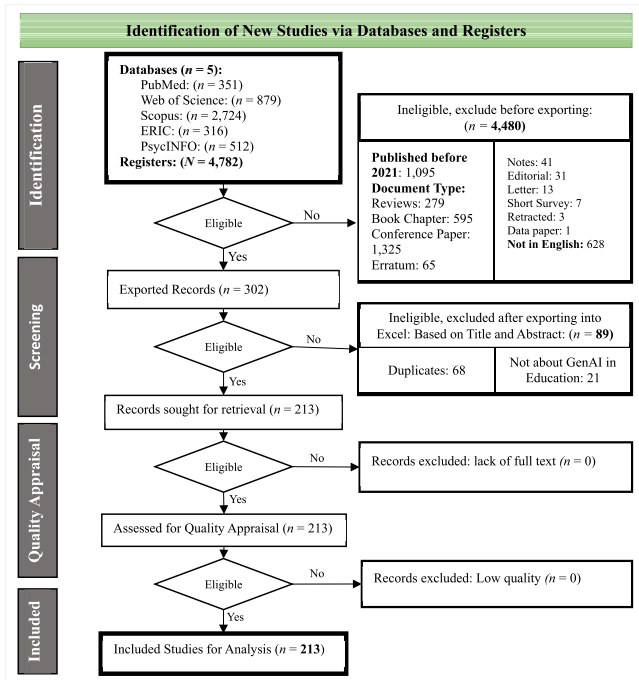


FIGURE 2 | PRISMA flowchart for the Sampling procedure.

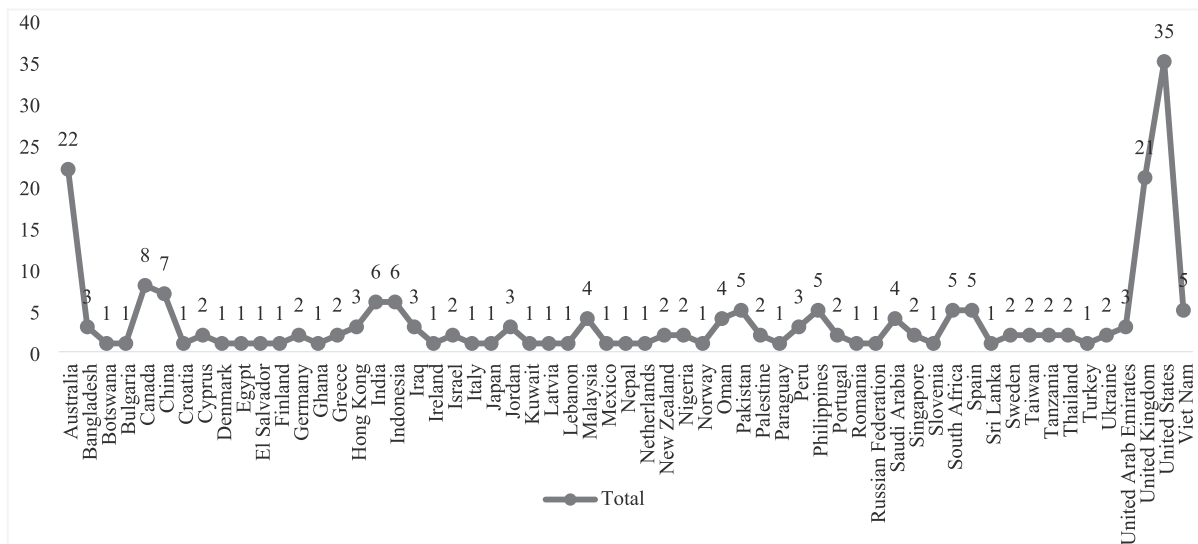


FIGURE 3 | Distribution of sample by country of affiliation ($n = 213$).

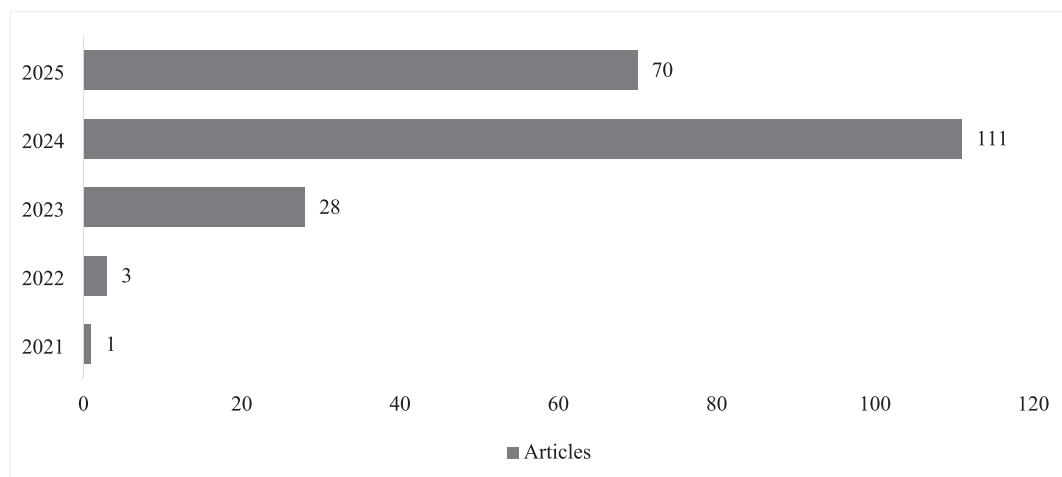


FIGURE 4 | Sample description by publication year.

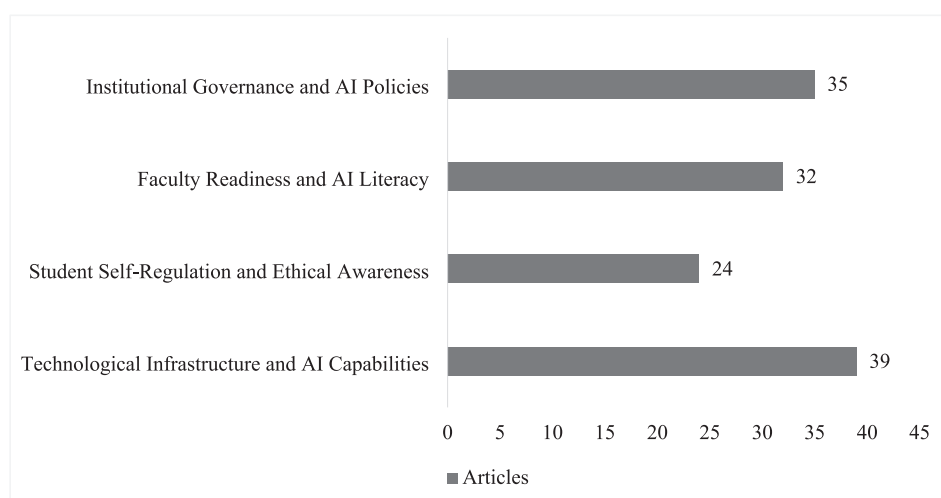


FIGURE 5 | Thematic factors influencing ethical integration of GenAI tools.

coursework evaluations. Additionally, Al-Jaghoub et al. (2025) and Pikhart and Al-Obaydi (2025) suggested that there were institutional biases in GenAI literacy training and availability, highlighting that STEM-focused faculties receive more GenAI integration support than humanities and social sciences departments. This disparity creates inconsistencies in GenAI governance, and it affects the uniformity of students' learning experiences.

Thus, to ensure responsible GenAI integration in the teaching and learning process, institutions should develop organised GenAI literacy programmes, embedding them within the faculty training frameworks. This will ensure that educators are equipped with practical GenAI ethical strategies and adaptive teaching methodologies. The universities should prioritise faculty engagement through ongoing GenAI ethics workshops, interdisciplinary GenAI literacy collaborations and professional development incentives to support faculty adaptation to GenAI-assisted education (Lund et al. 2025; Williams and Fadda 2023). Additionally, learning institutions should design customisable GenAI ethics modules aligned with disciplinary learning objectives to sustain a balanced approach to GenAI pedagogy.

4.3.1.3 | Student Self-Regulation and Ethical Awareness. The other thematic results of the included studies ($n = 25$) were categorised on the students' self-regulation and ethical awareness. The results indicated that ethical literacy frameworks should guide students' use of GenAI tools to minimise the risks of unintentional plagiarism and overreliance on GenAI-generated content. Findings by Abdi et al. (2025), Bayraktar Balkir and Zehir Topkaya (2025), and Michel-Villarreal et al. (2023) show that students struggled with GenAI transparency and unclear understanding of how GenAI-generated responses impact academic integrity. According to Nam and Bai (2023) and Rasul et al. (2023), the students who use GenAI tools without adequate critical evaluation training were more likely to submit GenAI-generated content without modification and contribute to the institutional concerns over academic dishonesty and misconduct.

Additionally, although GenAI literacy programs have been implemented in some cases, some disparities remain in student awareness of GenAI's ethical considerations, especially in institutions that have not fully integrated GenAI ethics into curricula (Katsantonis and Katsantonis 2024; Qu and Wang 2025). Similarly, Moorhouse et al. (2023) and Yusuf et al. (2024) indicate

that GenAI-enhanced learning environments require adaptive academic integrity measures, encouraging students to critically assess GenAI-generated materials instead of relying on automated responses. Universities should implement student-centred GenAI literacy frameworks, embedding ethical guidelines into coursework to promote self-regulated learning strategies that enhance intellectual accountability and critical thinking.

Thus, institutions may design GenAI literacy programs that are focused on promoting transparency, originality and self-regulation to improve ethical GenAI use. Indeed, Acosta-Enriquez et al. (2024) and Alshamy et al. (2025) proposed that GenAI-assisted learning should incorporate discussion-based assignments where students actively reflect on GenAI-generated suggestions to align them with scholarly and academic integrity principles. Furthermore, GenAI ethics training should be introduced as part of student orientation programs to promote early awareness of institutional GenAI integrity policies and equip students with the knowledge to navigate GenAI tools responsibly.

4.3.1.4 | Technological Infrastructure and GenAI Capabilities. The effectiveness of GenAI-integrated learning environments based on the sampled studies ($n=39$) depends on the quality of the technological infrastructure, particularly the accuracy of GenAI detection tools in identifying GenAI-generated content. Creighton et al. (2025), Liu et al. (2025) and Zhao, Cox, and Chen (2025) highlighted that AI-generated plagiarism detection tools have low reliability, producing false positives and negatives that complicate academic integrity assessments. Nikolic et al. (2025) and Perkins, Furze, et al. (2024) stated that the institutions that are employing GenAI-powered plagiarism detection models are often struggling to distinguish between human-authored submissions

and GenAI-assisted writing, which results in unfairness, especially in grading discrepancies.

Hence, Heugh et al. (2022) and Matthews and Volpe (2023) suggested that to improve GenAI-assisted assessments, universities should invest in adaptive GenAI detection frameworks, ensuring that GenAI-generated plagiarism checks align with evolving GenAI advancements. GenAI detection models should integrate multi-layered verification mechanisms, combining GenAI-generated response recognition with faculty monitoring in evaluations to improve assessment accuracy (Streletska et al. 2024; Sun, Zhao, and Li 2025). Moreover, GenAI-assisted learning tools must prioritise cognitive augmentation rather than knowledge replacement, ensuring GenAI serves as an aid to critical reasoning instead of an alternative to intellectual engagement. Figure 5 presents a summary of the synthesised result for RQ1.

Therefore, Figure 6 indicates that the sampled studies ($n=131$) emphasise that four interconnected factors shape the ethical integration of GenAI tools in higher education, that is, (i) the institutional governance models ($n=35$), (ii) faculty readiness and GenAI literacy ($n=32$), (iii) student self-regulation and ethical awareness ($n=25$), and (iv) technological infrastructure ($n=39$).

4.4 | Impact of GenAI Use on Academic Integrity and Institutional Mitigation Strategies

The rapid adoption of GenAI tools in higher education has presented a complex interplay of academic integrity risks, institutional precautionary measures, and the challenges in GenAI detection. This synthesis of empirical studies ($n=74$) provides results for RQ2 summarised in Table A2, on the impact of using GenAI tools on academic honesty, mitigation strategies, and

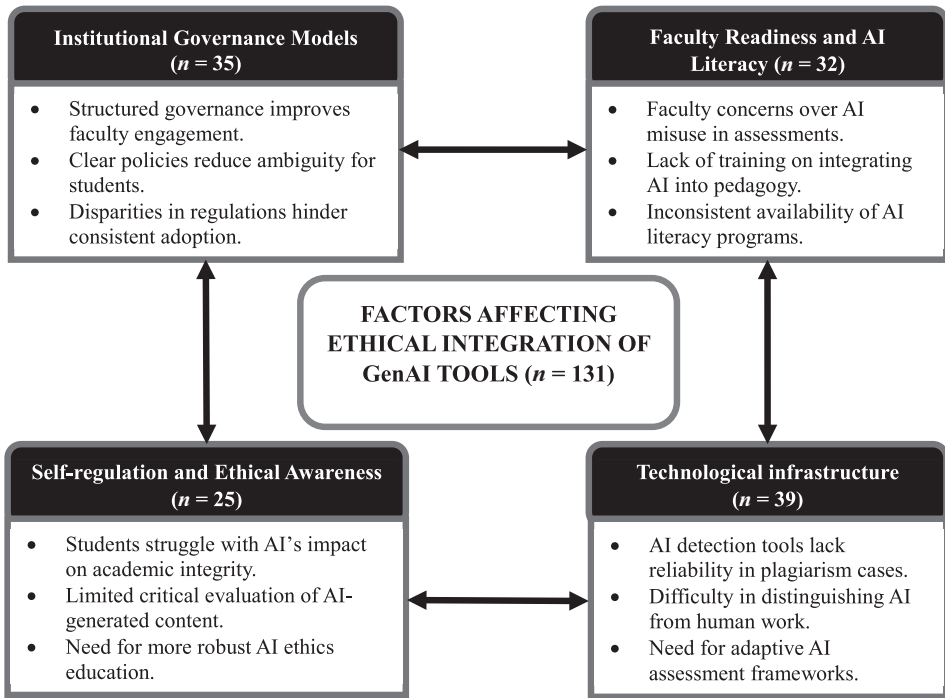


FIGURE 6 | Summary of factors affecting ethical integration of GenAI tools ($n=131$).

policy development for responsible and effective GenAI use as distributed in Figure 7.

4.4.1 | Academic Integrity Risks and AI-Assisted Misconduct

The use of Generative AI tools in higher education has introduced complex ethical dilemmas concerning academic integrity. Among the sampled studies ($n=39$), the increasing prevalence of GenAI-generated content is highlighted, which is undetectable by traditional plagiarism detection methods. This raises concerns about authorship accountability and transparency (Chan 2025a, 2025b; Ekaterina et al. 2025; Kofinas et al. 2025). Indeed, Abdi et al. (2025), Buniel et al. (2025) and Qu and Wang (2025) indicated that students are struggling to distinguish GenAI-generated materials from their work, and this increases the risk of unintentional academic misconduct. Additionally, Nikolic et al. (2023) and Williams and Fadda (2023) emphasised that without structured GenAI literacy programmes, universities reported higher incidences of student overreliance on GenAI tools for assignments, which potentially undermines their independent reasoning and development of critical thinking skills.

Moreover, Crawford et al. (2023), Hargreaves (2023) and Plata et al. (2023) identified that problems arise in subjects that rely heavily on written composition and analytical reasoning for assessments, where GenAI-generated paraphrasing and fabricated citations make plagiarism detection more challenging. Hence, traditional plagiarism detection software proves largely ineffective against GenAI-assisted modifications and creates gaps in academic integrity enforcement (B. W. Stone 2024). Indeed, fabricated references and citations generated by GenAI tools increase the risks of misinformation and may lead students to unknowingly submit falsified sources (Currie et al. 2023; Naidu and Sevnarayan 2023). Hence, Striepe et al. (2023) suggested that, to mitigate these ethical risks, institutions should revise their academic integrity policies to explicitly address GenAI-assisted research practices by incorporating GenAI literacy training into student orientations and coursework. Additionally, Malik and Terzidis (2025)

proposed adopting adaptive assessment models that prioritise discussion-based evaluations and oral reasoning exercises to reduce GenAI-induced academic dishonesty.

4.4.1.1 | Institutional Safeguards for Academic Integrity. The other set of the sampled studies ($n=19$) thematically indicated that the institutional mitigation strategies are fundamental in addressing the academic integrity challenges associated with the use of GenAI tools in education. Murch et al. (2025), Symeou et al. (2025), and Zlotnikova et al. (2025) emphasised the importance of GenAI ethics training programs to enhance the faculty's preparedness; however, the inconsistencies in the training accessibility reduced the efficacy of these interventions. Hence, Evangelista (2025) and Jin et al. (2025) found that the standardised GenAI literacy programs were underdeveloped in many institutions and led to disparities in faculty competency when addressing GenAI-assisted academic misconduct. Although GenAI-inclusive assessment designs are being introduced as a mitigation strategy, Almassaad et al. (2024), Plata et al. (2023) and Wilson (2025) indicated that faculty members require continuous professional development to enable them to seamlessly integrate GenAI into coursework and preserve academic integrity.

Moreover, to strengthen institutional governance in AI adoption, Katsamakos et al. (2024) and Teremetskyi et al. (2024) suggested that universities should embed GenAI ethics training into professional development frameworks, so that faculty are equipped with standardised strategies for detecting GenAI misuse and adapting assessment formats accordingly. Bikanga Ada (2024) and Driessens and Pischetola (2024) highlighted the necessity of establishing cross-institutional GenAI governance frameworks to facilitate collaborative policy implementation and academic integrity monitoring. Despite these efforts, Currie et al. (2023) and Jiang and Huang (2022) pointed to the persisting gaps in longitudinal studies to track the institutional adaptation to curb the GenAI-driven integrity risks and limited empirical validation of the long-term effectiveness of current mitigation strategies. Hence, they suggested that universities should address these gaps by forming interdisciplinary GenAI ethics committees, conducting longitudinal studies on GenAI

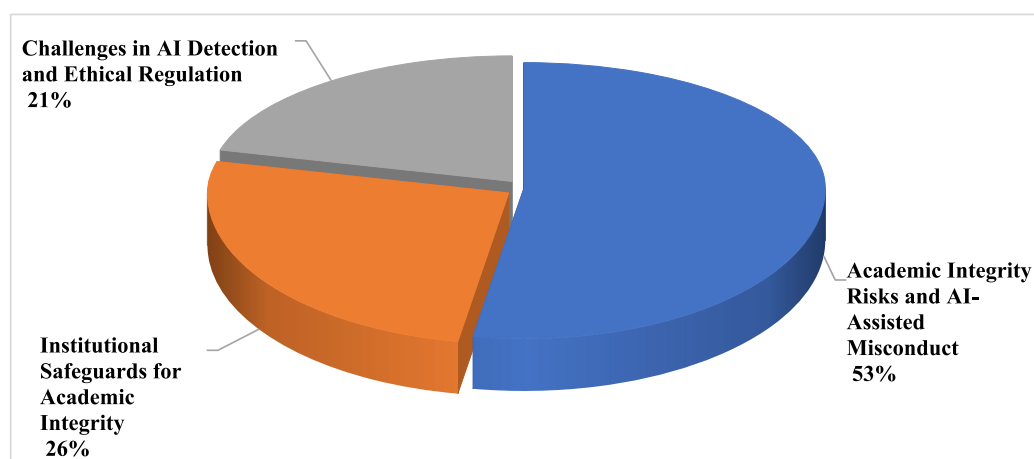


FIGURE 7 | Impact of GenAI adoption on academic integrity and institutional mitigation strategies ($n=74$).

adaptation and refining their governance models so as to align the evolving GenAI technologies with academic integrity standards.

4.4.1.2 | Challenges in GenAI Detection and Ethical Regulation. The efficacy of GenAI-powered plagiarism detection mechanisms was identified by the sample ($n=16$) to be a critical challenge in academic integrity enforcement. Almaghaslah (2025), Creighton et al. (2025), and Farber (2025) highlighted that a high rate of false positives and negatives undermined the significant reliability concerns regarding automated GenAI detection tools. Hence, faculty members were struggling to distinguish human-authored work from AI-generated submissions, especially in disciplines that require long-form written analysis and research-based assessments (Laflamme and Bruneault 2025; Velez and Rister 2025; Zhao, Cox, and Chen 2025). Furthermore, Klyshbekova and Abbott (2024) and J. Liu (2024) indicated that existing GenAI detection algorithms frequently misclassified paraphrased content, worsening the assessment discrepancies and fairness concerns. Additionally, Bellini et al. (2024), Slimi et al. (2025) and Slimi and Villarejo-Carballido (2024) recommended the implementation of hybrid GenAI-human assessment frameworks, improving detection reliability by integrating manual monitoring alongside GenAI-powered plagiarism detection models. The other studies by Bannister et al. (2024) and Elbaz et al. (2024) proposed that the institutions should also invest in refining GenAI detection algorithms tailored to disciplinary-specific needs to promote accuracy in content authenticity verification. Similarly, Bouteraa et al. (2024) and Shabbir et al. (2024) suggested that universities require comprehensive GenAI ethics frameworks to be embedded into curricula and faculty training programmes to ensure responsible AI adoption. However, limited empirical research exists on GenAI detection reliability across diverse academic disciplines, making the development of scalable integrity enforcement models a pressing priority (Bikanga Ada 2024; Tang et al. 2024). Indeed, to mitigate these issues, institutions need to refine GenAI detection mechanisms, develop hybrid assessment models, and implement interdisciplinary GenAI ethics training to strengthen academic integrity across educational domains as summarised in Figure 8 below.

Integrating GenAI tools in higher education greatly influences academic integrity, presenting ethical risks and institutional responses to mitigate these challenges, as evidenced by the sampled studies ($n=74$). The findings by Chan et al. (2025), Ekaterina et al. (2025) and Kofinas et al. (2025) revealed that GenAI-generated content often evades traditional plagiarism detection, increasing concerns over authorship transparency and academic dishonesty. Similarly, Abdi et al. (2025), Buniel et al. (2025) and Qu and Wang (2025) discovered that students face difficulties in differentiating GenAI-generated content from their intellectual contributions, which can cause the risk of academic misconduct. Similarly, Murch et al. (2025) and Zlotnikova et al. (2025) found that although institutions are developing GenAI ethics training initiatives, variations in faculty preparedness continue to limit their widespread effectiveness. Moreover, Evangelista (2025), Jin et al. (2025) and Rana (2025) found that universities also implemented adaptive assessment models and cross-institutional governance frameworks to align GenAI literacy with ethical academic practices. However, Creighton et al. (2025) and Liu et al. (2025) stated that GenAI detection tools are still unreliable and undermine the plagiarism enforcement mechanisms. Thus, Bikanga Ada (2024) and Tang and Eaton (2024) suggested that to strengthen integrity protections, the GenAI detection algorithms should be refined by integrating the blended GenAI-human assessment models and embedding GenAI ethics education within curricula, to ensure responsible GenAI adoption and preserve academic integrity. Indeed, these findings highlight the changing landscape of GenAI tools in education and how institutions should continuously adapt policy frameworks to promote the balance between technological innovation and academic standards.

4.5 | Model Practices for Ethical and Effective GenAI in Teaching and Learning

The synthesis of the results of the sampled studies ($n=114$) for the third research question (RQ3 as in Table A3) highlights the model practices for faculty training, student-GenAI engagement, and GenAI-inclusive assessment models. It offers actionable insights for the responsible use of GenAI tools to mitigate

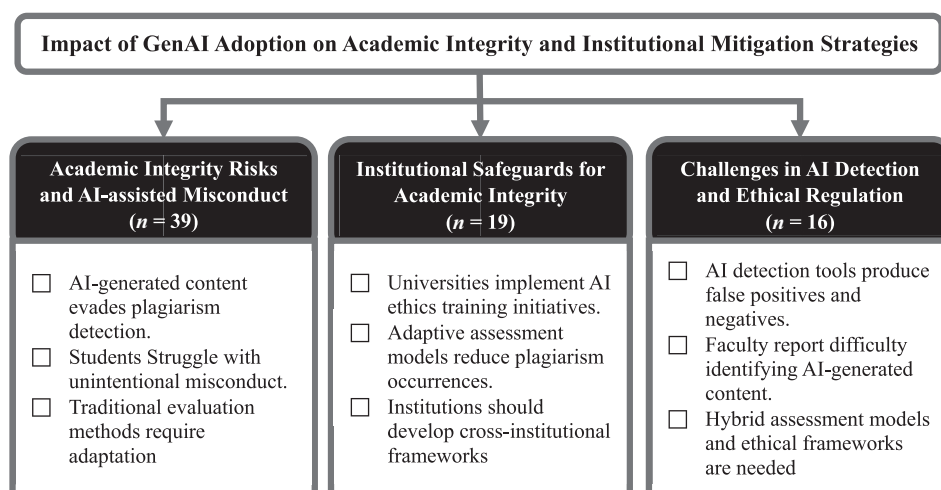


FIGURE 8 | GenAI adoption on academic integrity and institutional mitigation strategies ($n=74$).

academic integrity risks. Figure 9 presents the thematic distribution of the sample.

4.5.1 | Faculty-Led GenAI Literacy and Ethical Training

The thematic subcategorisation of the sample ($n=38$) indicated that the use of GenAI tools in higher education requires the faculty members to be more professionally prepared to ensure ethical and practical pedagogical alignment in their use. Findings by Alqahtani and Wafula (2025), Nguyen et al. (2025) and Symeou et al. (2025) show that the uncertainties surrounding GenAI-assisted pedagogy primarily stem from inadequate AI literacy training. Additionally, Pikhart and Al-Obaydi (2025) and Wilson (2025) pointed out that integrity workshops can significantly improve faculty confidence in adopting GenAI-integrated teaching strategies. However, Nikolic et al. (2023), Perkins (2023), Perkins and Roe (2024b) and Sullivan et al. (2023) also identified the inconsistencies in participation rates and accessibility as the main hindrance to the usefulness of the workshops. They noted that STEM disciplines receive more structured GenAI training support than other disciplines, such as humanities and social sciences. Moreover, Hammond et al. (2023), Pham et al. (2023) and Williams and Fadda (2023) suggested that institutions may address the training disproportions by integrating mandatory GenAI literacy programs within the faculty's continuous professional development initiatives, focusing on equitable access to GenAI ethics training across disciplines. The studies underscore the need for interdisciplinary training frameworks that bridge the use of GenAI tools in STEM and non-STEM fields. However, a significant challenge is that there is low empirical validation of faculty-led GenAI ethics training programs. Donnelly and Kennelly (2024), Drumm and Sami (2024), Gabriel (2024), Miltenoff (2024), Naidu and Sevnarayan (2023) and van Rensburg and van der Westhuizen (2024) underline the insufficient cross-institutional comparisons of GenAI literacy curricula as the limiting factor against the generalisable model pedagogical practices for GenAI use in teaching.

Furthermore, to refine faculty preparedness, Alexander et al. (2023), Bozkurt (2024), Currie and Barry (2023) and Zlotnikova et al. (2025) proposed that institutions should conduct GenAI ethics workshops by integrating them into

tenure-track and continuing professional development programmes. Additionally, Chou and Lee (2022), Migliorini and Moreira (2025), Miltenoff (2024), Nam and Bai (2023) and Rasul et al. (2023) suggested that cross-institutional faculty collaborations can help in developing disciplinary-specific GenAI literacy models that standardise GenAI training for educators across various academic contexts. Hence, the standardisation would ensure that educators are equipped with practical knowledge to effectively integrate GenAI tools and promote academic integrity. Indeed, educators should be equipped with all the necessary GenAI literacy and ethical guidelines to responsibly integrate GenAI tools into pedagogical methodologies. These GenAI literacy and responsible integration strategies are summarised in Table 6 below.

4.5.2 | Student-Centred GenAI Literacy and Responsible GenAI Use

The included studies ($n=39$) showed that students face numerous challenges in understanding the ethical implications of GenAI adoption, with Abdi et al. (2025), Alshamy et al. (2025) and Bayraktar Balkir and Zehir Topkaya (2025) highlighting major concerns as GenAI transparency, unintended plagiarism, and overreliance on GenAI-generated content. Buniel et al. (2025), Qu and Wang (2025) and Uppal and Hajian (2025) demonstrate that the universities that embedded GenAI ethics training within curricula observed improved student self-regulation and critical engagement with GenAI-assisted learning. However, despite the benefits of GenAI-enhanced learning environments in higher education, other studies, such as Perkins and Roe (2024a), Sullivan et al. (2023) and Zhao, Chen, and Cox (2025) suggest that students need structured academic integrity safeguards to guarantee the responsible use of GenAI tools.

Hence, to promote academic integrity in the use of GenAI tools, Donnelly and Kennelly (2024), Nikolic et al. (2023), van Rensburg and van der Westhuizen (2024), A. Williams (2024) and P. Williams (2023a) suggested that institutions of learning should design adaptive GenAI literacy frameworks that are well aligned with the students' GenAI learning experiences. Additionally, Crawford et al. (2023), Hsiao and Tang (2024) and Zlotnikova et al. (2025) also emphasised the importance of GenAI integrity orientation programmes, especially for

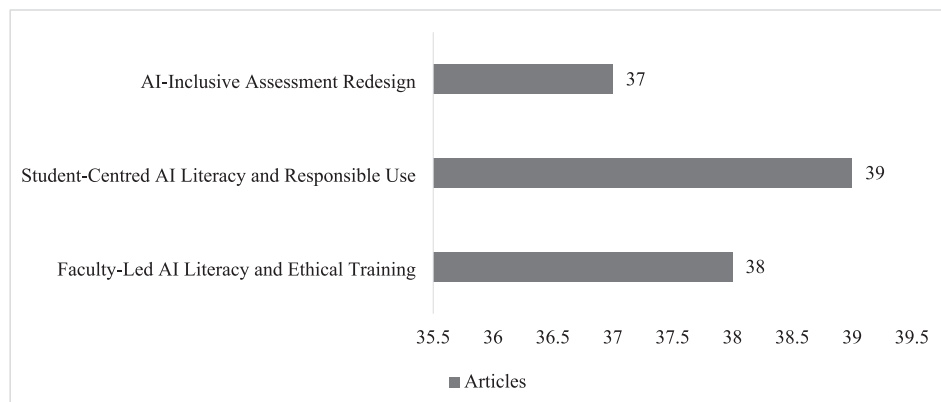


FIGURE 9 | Thematic model practices for ethical and effective GenAI in teaching and learning.

TABLE 6 | GenAI literacy and responsible integration strategies.

Faculty-led AI literacy component	Practical integration strategy
Standardised AI Ethics Training	Embed mandatory GenAI literacy programs within faculty development initiatives to mitigate uncertainty in AI-assisted pedagogy
Interdisciplinary AI Training Models	Develop cross-disciplinary GenAI literacy frameworks to bridge STEM and non-STEM applications
Tenure-Integrated AI Ethics Workshops	Incorporate GenAI ethics workshops within tenure-track and professional development programs to refine AI literacy
Cross-Institutional AI Literacy Models	Foster faculty collaborations to standardise AI ethics training across multiple institutions, promoting consistency in pedagogical approaches
Empirical Validation of AI Training Programs	Conduct comparative studies on faculty AI ethics curricula to refine best practices and address gaps in institutional training

first-year students, to instil ethical GenAI practices early in their academic journey. On the other hand, despite these interventions, Plata et al. (2023) and A. Stone (2023) revealed a lack of longitudinal research examining how students adapted to GenAI-mediated learning environments over time. Similarly, Currie and Barry (2023) and Naidu and Sevnarayan (2023) observed that student-centred GenAI ethics frameworks remain underdeveloped in higher education, with limited empirical validation of their effectiveness for different learning styles.

Furthermore, Alexander et al. (2023), Alexander et al. (2025) and Malik, Pratiwi, et al. (2023) encouraged the learning institutions to develop interactive GenAI literacy modules to provide students with case-based learning experiences and help them to critically evaluate GenAI content for accuracy and other integrity issues. Indeed, Alderman et al. (2025), Anurag and Johnpaul (2024), Liu, Sullivan, and Sinnamon (2024), Matich and Mousannif (2024) and Osifo (2023) concluded that if GenAI transparency case studies are integrated into academic integrity training, they would equip students with practical strategies for distinguishing GenAI-generated content from original work. Additionally, Rasul et al. (2024) and Whitfield and Hofmann (2023) indicated that implementing GenAI-assisted critical engagement assignments may also strengthen the students' ability to use GenAI tools more responsibly while promoting their intellectual accountability and independent reasoning. Hence, enhancing the students' understanding of ethical GenAI usage, promoting critical engagement and academic integrity as summarised in Table 7.

TABLE 7 | GenAI usage to promote critical engagement and academic integrity.

Student-centred GenAI literacy component	Practical integration strategy
Adaptive GenAI Literacy Frameworks	Align GenAI ethics education with students' learning experiences to reinforce responsible engagement
Early GenAI Integrity Orientation	Introduce structured GenAI ethics training for first-year students to instil foundational ethical guidelines
Case-Based GenAI Transparency Modules	Implement applied ethics case studies where students critically assess GenAI-generated content
GenAI-Assisted Critical Engagement Assignments	Develop assignments requiring students to analyse GenAI-generated material, assess biases, and compare it with human-authored texts
Longitudinal Research on Student GenAI Adaptation	Conduct studies evaluating students' evolving engagement with GenAI-mediated learning environments

4.5.3 | AI-Inclusive Assessment Redesign

The use of GenAI tools in higher education has led to transformations in assessment design; however, the synthesised results ($n=37$) revealed that institutions were struggling to balance GenAI innovation with academic integrity. Studies such as Almaghaslah (2025), Evangelista (2025) and Kofinas et al. (2025) indicated that GenAI-integrated assessments have been shown to reduce academic misconduct risks, but challenges persist in distinguishing GenAI-generated responses from student-authored work. Hence, Chan et al. (2025), Creighton et al. (2025), T. Liu (2023) and Liu et al. (2025) highlighted that educators are facing difficulties evaluating GenAI-generated submissions due to inconsistencies in plagiarism detection algorithms, which produce false positives and negatives. Additionally, Fuad et al. (2024), Kodali et al. (2023), Leong and Zhang (2025) and Pudasaini et al. (2024) underscore the need for multi-layered verification models that account for GenAI-generated paraphrasing and automated responses.

Similarly, Donnelly and Kennelly (2024) and Khlaif et al. (2025) encouraged the institutions to refine GenAI-integrated assessment frameworks to ensure that GenAI plagiarism detection tools conform to grading mechanisms while preserving academic originality standards. Nevertheless, Al Harrasi and El Din (2024), Kortemeyer and Nöhl (2025) and Suhail et al. (2024) indicate insufficient research on faculty-student dynamics in GenAI-assisted grading, with limited empirical validation

of GenAI-inclusive assessments. Furthermore, Prajapati et al. (2025) and Pudasaini et al. (2024) emphasised the need for improved GenAI detection models that can reliably differentiate AI-generated and human-authored responses.

Moreover, institutions may implement hybrid GenAI-human evaluation models to enhance GenAI-inclusive assessment design, like Cebon and Doyle (2024), Le and Metzger (2024), Mo et al. (2022) and Naidu and Sevnanarayan (2023) suggested a combination of faculty review with GenAI-assisted plagiarism detection for improved academic integrity monitoring. The faculty-guided GenAI assignment reviews will ensure ethical AI-assisted grading practices as Ibrahim et al. (2023), Striepe et al. (2023) and Wang, Fang, and Wang (2024) show that this would allow educators to apply better evaluation criteria beyond automated detection mechanisms. Indeed, they advocated for multi-layered verification strategies to integrate GenAI plagiarism detection with faculty input to strengthen assessment reliability and safeguard academic integrity in GenAI-mediated coursework. Thus, Table 8 summarises the GenAI-supported assessment models on how to effectively integrate GenAI-supported assessments to uphold academic integrity and leverage GenAI's pedagogical potential.

Therefore, institutional regulation of the integration of GenAI tools in education requires a balanced approach that promotes both technological advancement and ethical responsibility. They should continuously develop explicit integrity policies, promote GenAI literacy and address ethical concerns through strong institutional integrity guidelines. This process would enable learning institutions to nurture the benefits of GenAI tools and effectively maintain academic integrity

TABLE 8 | AI-supported assessment models to uphold academic integrity.

AI-inclusive assessment redesign component	Strategic implications
Hybrid GenAI-Human Evaluation Models	Combine faculty monitoring with GenAI-assisted plagiarism detection to refine grading standards
Multi-Layered Verification Strategies	Enhance plagiarism detection mechanisms by integrating GenAI response analysis with faculty-driven evaluations
Refined GenAI Plagiarism Detection Tools	Ensure plagiarism detection algorithms align with grading mechanisms to safeguard originality
Faculty-Guided GenAI Assignment Reviews	Implement educator-led evaluations to apply discipline-specific integrity criteria to GenAI-assisted coursework
Empirical Research on GenAI-Assisted Grading	Expand research on faculty-student interactions in GenAI-mediated assessment practices

and create a supportive learning environment for all GenAI users (Bouteraa et al. 2024; James and Filgo 2023; McIntire et al. 2024; A. Stone 2023).

5 | Discussion

This section discusses the balance between innovation and ethics in using GenAI tools in higher education, with a particular focus on the role of institutional guidelines in enhancing academic integrity. It discusses the synthesised findings from 213 studies published in English between 1 January 2021 and 31 May 2025 on the impact of educational support, self-regulation and institutional regulations on the ethical use of GenAI tools. Results indicated that academic support, including resource availability, training, and technical guidance, is central to effectively integrating GenAI tools in higher education. Additionally, the findings highlighted the importance of self-regulation in maintaining academic integrity, with adaptive learning and personalised feedback playing significant roles. Similarly, institutional regulations, including data privacy, bias, and fairness guidelines, are vital in promoting responsible use and preventing academic malpractice. Thus, this discussion is organised according to the order and scope of the three research questions.

5.1 | Factors Influencing the Integrity of GenAI Integration in Higher Education

The empirical evidence from 2021 to 2025 revealed that four main interdependent factors critically shape the integrity of integrating GenAI tools in higher education. These factors include technological infrastructure, institutional governance, faculty preparedness and student self-regulation. The findings emphasise that academic integrity is not merely contingent upon individual behaviour but emerges from systemic design and implementation. Therefore, institutional guidelines are fundamental in promoting academic integrity. It implies that institutions with clear, enforceable GenAI policies, well-articulated through cross-disciplinary committees and ethics review boards, demonstrated improved transparency and coherence in implementing GenAI tools (Alqahtani and Wafula 2025; Johnson and Zhang 2024). Conversely, universities without strong regulatory frameworks reported more interpretive inconsistencies, mainly in assessing student submissions generated with GenAI (Currie et al. 2023; Henry and Oliver 2022). Indeed, the variations in policy maturity highlight the urgency for unified governance models that align with academic integrity norms while remaining agile to GenAI developments (Yang et al. 2024).

Additionally, faculty preparedness is another critical factor affecting the pedagogical legitimacy of using GenAI tools. The structured training in GenAI ethics and application increases educator confidence (Malik and Terzidis 2025), but disciplinary inequities continue. For instance, Al-Jaghoub et al. (2025), Streletska et al. (2024) and Tlais et al. (2025) found that STEM faculties benefited more from early integration and funding support. In contrast, in the humanities and social sciences, educators reported insufficient access to relevant tools or guidance.

Hence, this asymmetry compromised the equitable GenAI implementation and limited its transformative potential in non-STEM disciplines.

Furthermore, student self-regulation and GenAI literacy were underdeveloped across multiple academic contexts. The synthesised results suggested that students often misinterpret the GenAI outputs as inherently valid, resulting in uncritical reproduction of content and a breach of academic integrity (Nam and Bai 2023; Rasul et al. 2023, 2024). Particularly, institutions that embedded GenAI ethics modules into their curricula, such as through reflective assignments or scenario-based assessments, saw measurable improvements in the students' ethical reasoning and self-monitoring skills (Acosta-Enriquez et al. 2024; Moorhouse et al. 2023).

Moreover, technological infrastructure was also found to be pivotal in the integration process, but it remains a limiting factor. Similarly, Gupta and Gupta (2024), Leong and Zhang (2025) and Smith and Lester (2024) indicated that although automated GenAI detection tools are increasingly being adopted, they frequently yield false positives or overlook nuanced GenAI-generated content. Indeed, the learning institutions that used multi-layered verification by merging algorithmic detection and human judgement reported improved reliability in academic misconduct identification (Bellini et al. 2024; Fuad et al. 2024; Giray 2024). However, such solutions require sustained investment in adaptive learning analytics and faculty training to interpret algorithmic outputs ethically. Therefore, these findings suggest that a fragmentary approach to GenAI integration in the learning environment undermines academic integrity. Instead, institutions should pursue a well-coordinated GenAI governance framework that embeds GenAI literacy across stakeholders, disciplines and decision-making structures. Hence, a systems-based response to the factors that influence the integrity of integrating GenAI tools in education is critical for sustaining trust, fairness and intellectual rigour in GenAI-mediated education.

5.2 | Impact of GenAI Integration on Academic Integrity and Effective Institutional Mitigation

The integration of GenAI in higher education has presented both transformative opportunities and unprecedented academic integrity risks. According to Chan et al. (2025) and Kofinas et al. (2025), GenAI tools are enhancing educational accessibility and efficiency while simultaneously disrupting established norms of assessment reliability, academic authorship, and intellectual autonomy. Thus, Currie et al. (2023) and Katsantonis and Katsantonis (2024) advised that the institutions that adopted a reactive rather than proactive approach to GenAI integration reported higher instances of academic misconduct and confusion around ethical boundaries when using GenAI tools.

GenAI-assisted misconduct, though not always associated with intentional plagiarism, through overreliance on automated GenAI outputs, diminishes the students' engagement with evidence-based reasoning, source evaluation, and argument synthesis (Giray 2024; Leong and Zhang 2025; Zhang

et al. 2025). They submit algorithmically generated content as their original work without evaluating its epistemic validity and ethical implications. Similarly, Qu and Wang (2025) observed that institutions that lacked structured GenAI literacy training reported higher incidences of citation errors, shallow paraphrasing, and misrepresentation of sources. These patterns suggest that academic integrity violations are mainly a result of epistemological naivety rather than wilful academic dishonesty.

Moreover, faculty engagement is a critical factor in mitigating breaches of GenAI-related academic integrity. Nevertheless, Evangelista (2025) and Murch et al. (2025) highlighted the inconsistent faculty preparedness, especially across non-STEM disciplines. Hence, in the absence of standardisation of the GenAI-integrated instructional frameworks, many educators revert to traditional assessment models, which are ill-suited to GenAI-mediated learning contexts (Symeou et al. 2025). Bikanga Ada (2024) and Jin et al. (2025) noted that the introduction of GenAI-inclusive coursework, like scaffolded assignments that require human-GenAI collaboration, has the potential to preserve academic rigour and leverage GenAI's potential to enhance academic skills development. However, disparities in faculty access to GenAI pedagogical development are still a critical bottleneck in higher learning institutions.

Furthermore, the automated GenAI detection tools are flawed and yet are widely deployed as mitigation strategies for academic integrity risks. Indeed, despite institutional enthusiasm, these detection tools are often characterised by poor interpretability and false positives, unduly affecting students (Klyshbekova and Abbott 2024; Zhao, Cox, and Chen 2025). For instance, Laflamme and Bruneault (2025) revealed that algorithmic detectors misclassified 23% of human-authored essays and unreasonably penalised students who used uncommon rhetorical writing styles. Hence, Xu, Zhan, and Liang (2024) and Xu and Jumaat (2024) suggested that mitigation strategies should move toward hybrid verification models that combine contextual human judgement with algorithmic outputs.

Additionally, Alshamy et al. (2025) and Moorhouse et al. (2023) established that the institutions that adopted anticipatory models, such as including GenAI literacy in academic skills modules and framing GenAI-assisted assignments within ethical discussion, had lower rates of academic misconduct and stronger student learning engagement. Moreover, interdisciplinary faculty development programmes that include scenario-based teaching, GenAI ethics training, and critical evaluation approaches were more effective in supporting faculty confidence and instructional consistency (Driessens and Pischetola 2024; Teremetskyi et al. 2024). Indeed, Elbaz et al. (2024) concluded that the cross-sector collaborations between universities, technologists, and librarians were particularly impactful in creating adaptive governance frameworks that keep pace with evolving GenAI capabilities.

Therefore, the integration of GenAI tools in higher education reshapes academic integrity. Rather than depending exclusively on punitive investigation mechanisms, institutions

should reframe the use of GenAI tools as a pedagogical opportunity to nurture epistemic agency, ethical literacy, and critical inquiry. The effective mitigation should be based on multidimensional strategies that include standardised GenAI literacy curricula, faculty professional development, and automatic detection protocols that are sensitive to disciplinary distinctions. Hence, sustaining academic integrity in GenAI-mediated environments requires moving beyond compliance toward a culture of informed and principled GenAI use in education.

5.3 | Educational Evidence-Based Model Practices for Ethical and Effective GenAI Integration

The ethical deployment of GenAI tools in higher education hinges not only on governance structures but also on the pedagogical sophistication of faculty training and the depth of student AI literacy. The evidence supports a systems-based approach. The one that embeds interdisciplinary collaboration, dynamic policy feedback, and tailored pedagogical design to safeguard academic integrity to maximise the effective use of GenAI tools.

5.3.1 | Institutional GenAI Governance Models

The synthesised results show that the institutions that implemented interdisciplinary GenAI Ethics Committees, with representation from technology, pedagogy and academic policy, demonstrated higher responsiveness to GenAI-related misconduct and stronger stakeholder confidence in policy legitimacy (Hegazy et al. 2024; Lentz 2024; Zlotnikova et al. 2025). However, without iterative policy evaluation, it proved insufficient. Indeed, static policies rapidly become obsolete as GenAI capabilities evolve fast, exposing the gaps in enforcing academic integrity. The model practices emphasise cyclical policy improvement, in which the feedback from GenAI implementation is used to revise the GenAI guidelines regularly and create adaptable governance structures attuned to empirical shifts.

5.3.2 | Faculty Development as an Ethical and Disciplinary Imperative

The most effective faculty training programs should incorporate GenAI tool proficiency and promote source verification, critical engagement with algorithmic bias, and discipline-specific ethics (Alderman et al. 2025; Jain and Menon 2023; Sullivan et al. 2023). Remarkably, faculty in STEM fields benefited from workshops on GenAI-augmented assessment accuracy. However, humanities and social sciences faculties required support in evaluating GenAI-generated claims and integrating misinformation screening into coursework (Jaidka et al. 2025; Nam and Bai 2023). Institutions that integrated GenAI ethics into faculty promotion and professional development standards reported more consistent ethical engagement across departments (Elom et al. 2025; Farrell 2024; Mondal 2025). Indeed, generalised training models failed to account for contextual pedagogical challenges, reducing faculty agency and risking instructional inequity. Indeed, faculty collaborations emerged as model practices to facilitate knowledge exchange and promote

disciplinary customisation. These academic and professional networks would help to democratise access to GenAI teaching strategies and help educators align localised teaching needs with global ethical standards.

5.3.3 | Student-Centric GenAI Literacy and Active Integrity Frameworks

The GenAI literacy efforts targeting students were most effective when structured as ongoing and inquiry-driven programmes rather than isolated workshops. Hence, the institutions that deployed interactive case-based modules, allowing students to compare GenAI- and human-authored content, fostered improvement in students' ability to identify misinformation and question algorithmic reliability (Elom et al. 2025; Farrell 2024; Mondal 2025). Moreover, introducing the longitudinal tracking mechanisms, such as the semester-based GenAI usage audits and reflective journals, allowed educators to detect changes in student reliance on GenAI content and tailor interventions accordingly. Indeed, Evangelista (2025) and Kofinas et al. (2025) suggested that reframing academic integrity training toward intellectual autonomy rather than surveillance and punishment encourages deeper ethical reasoning.

Additionally, GenAI-inclusive assessments requiring students to refine GenAI outputs or critique algorithmic decisions demonstrated the benefits of promoting critical thinking and minimising passive GenAI dependence (Rasul et al. 2023; Zhao, Cox, and Chen 2025). Integrating GenAI tools in teaching and learning demands more than procedural compliance. It requires institution-wide cultural and pedagogical adaptation. Therefore, the evidence-based best practices include establishing interdisciplinary ethics committees with iterative oversight, tailoring faculty training to technical and disciplinary needs, and designing student literacy programmes that foster epistemic agency. It implies that ethical GenAI use develops not through prohibition but through deliberate educational design that empowers students and educators to navigate GenAI responsibly, critically, and reflectively.

5.3.4 | Standardising GenAI Integrity Guidelines Across Educational Systems

The perceived policy fragmentation remains a significant vulnerability in GenAI governance. For example, universities in high-tech regions tend to offer comprehensive GenAI ethics modules, while under-resourced institutions struggle to implement basic GenAI literacy instruction (Gabriel 2024; Miltenoff 2024; van Rensburg and van der Westhuizen 2024). Indeed, unreliable plagiarism detection further compounds this gap and unfairly evaluates GenAI-generated outputs across institutions (Leong and Zhang 2025).

Therefore, to address these concerns, this literature review advocates for GenAI integrity frameworks that balance uniform benchmarks with disciplinary flexibility. These frameworks should outline essential competencies in GenAI use, faculty engagement, and academic assessment and accommodate field-specific norms, such as interpretive analysis in the humanities

against algorithmic verification in STEM (Giray 2024; Wang, Huai, et al. 2024). Similarly, the policy examples from GenAI-intensive regions suggest that adaptive governance models driven by empirical review cycles enable institutions to update ethics protocols in response to technological advancements and pedagogical needs (Kangwa et al. 2025). Thus, policy coherence promotes transparency and trust among learners and educators, the essential pillars for responsible GenAI integration.

5.3.5 | Ethical Alignment in GenAI Model Design

The ethical and pedagogical relevance of GenAI tools is shaped by their architecture. Hence, the best practices underscore the importance of developer-institution collaboration to ensure that GenAI tools are built with features that align with academic integrity standards (Agarwal and Sivaraman 2024; Corbeil and Corbeil 2025; Valdivieso and González 2025). For example, the models should include built-in bias alerts, citation auditing, and discipline-specific logic flags to support critical engagement in pedagogical settings (X. Liu 2025; Plata et al. 2023). If these features are not integrated, they risk producing GenAI tools that reinforce dominant epistemologies while marginalising alternative ways of knowing, a serious concern for inclusivity in the design (A. Stone 2023). Hence, GenAI developers must transition from generalised usability goals to context-sensitive ethical designs tailored to educational environments through participatory development with stakeholders in teaching, learning, and policy domains.

Therefore, effective GenAI adoption is not reducible to institutional innovation alone. Rather, it requires a multilevel

ecosystem of evidence-informed strategies. It entails standardising the ethical frameworks to ensure equitable governance and reengineering GenAI tools, embedding pedagogical integrity at their core. Indeed, without investment in policy and design, the potential of GenAI to advance educational quality will remain unevenly distributed and ethically unstable.

Figure 10 illustrates that GenAI's ethical and effective use in education requires integrating the contributions of multiple stakeholders, including policymakers, institutional leaders, faculty, students and GenAI developers. Integrating GenAI tools in education without cross-sector collaboration risks reinforcing systemic inequities, academic integrity concerns and inadequate pedagogical alignment. Hence, institutions should refine GenAI literacy initiatives, policymakers should standardise GenAI ethics policies and technology developers should prioritise ethical model design to ensure that GenAI tools support responsible learning across educational environments.

5.4 | Limitations and Recommendations for Future Research

This systematic review adhered to the PRISMA framework and applied a rigorously defined set of inclusion and exclusion criteria (see Table 2), which included 213 peer-reviewed articles for analysis. Indeed, the study exclusively sampled peer-reviewed, full-text research articles indexed in five major scholarly databases (PubMed, Scopus, Web of Science, ERIC and PsycINFO). Although this approach ensured the quality and relevance of the included studies, it may have excluded high-quality, non-English

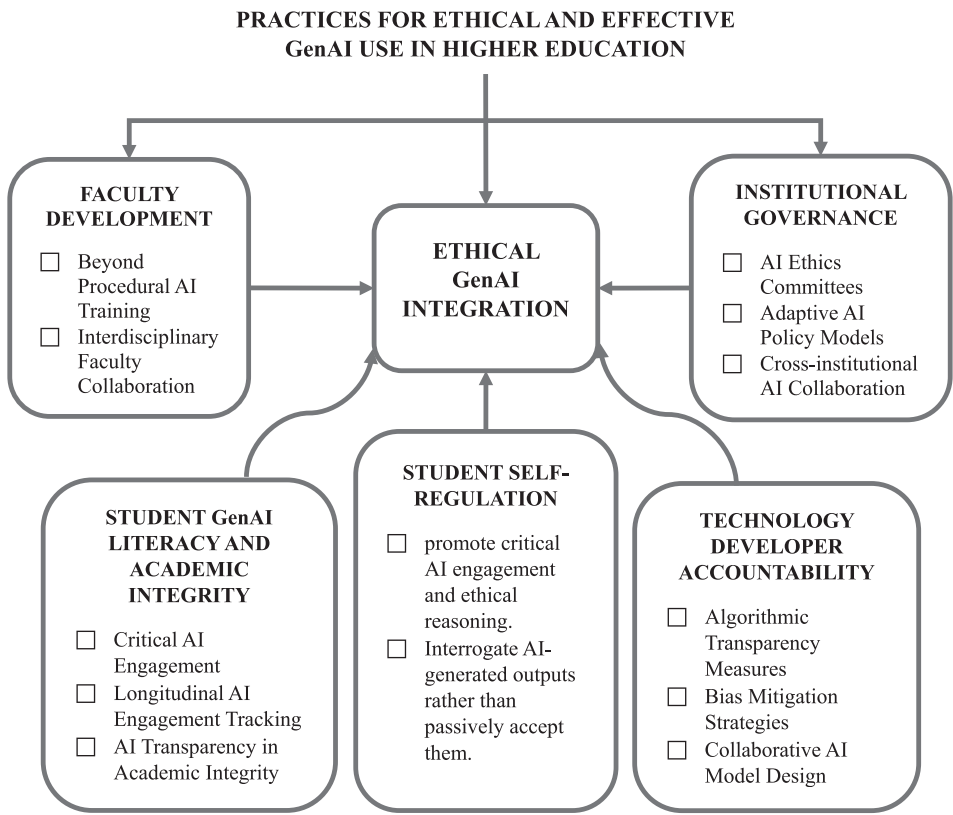


FIGURE 10 | Evidence-based best practices for ethical and effective GenAI adoption.

publications or grey literature, which could have limited the multilingual diversity of perspectives on GenAI integration in higher education. This potential language and indexing bias may particularly underrepresent insights from low- and middle-income countries, where publication visibility in indexed databases is more constrained.

Additionally, the temporal restriction to articles published between January 1, 2021, and May 31, 2025 ensured focus, although this may have excluded recent studies on rapidly evolving GenAI technologies published post-inclusion period. The rapid GenAI development and deployment in higher educational contexts implies that some insights captured in the review's earlier years may no longer reflect current technological capabilities, ethical concerns or pedagogical practices. Hence, while temporal delimitation was necessary to ensure analytical focus, it also introduces limitations in addressing the ongoing evolution of GenAI tools.

Therefore, these limitations underscore the need for future systematic reviews to extend their scope by incorporating multilingual and non-indexed databases, but with a strengthened quality appraisal process. Future research may apply mixed-methods synthesis strategies and adopt adaptive inclusion criteria that account for the changeability of GenAI technologies in education. This may help to triangulate current findings and improve their applicability in diverse global higher education settings.

6 | Conclusion

This study critically investigated the intersection of academic integrity and the integration of generative artificial intelligence in higher education. It was a synthesis of research evidence from 213 peer-reviewed results that were published between 2021 and 2025. The systematic literature review was framed by the Technology Acceptance Model (TAM), the Theory of Planned Behaviour (TPB), and Self-regulated learning theory to address the three guiding research questions concerning the factors influencing GenAI integration (RQ1), its impact on academic integrity (RQ2), and evidence-based practices for its ethical and pedagogical use (RQ3).

The findings underline that academic support mechanisms, including structured training and institutional resource availability, play a pivotal role in shaping students' acceptance and ethical use of GenAI tools (Lentz 2024; Nguyen 2025; Tang et al. 2024). In alignment with TAM, the perceived usefulness and ease of use of GenAI were found to be positively associated with students' willingness to employ these tools to develop academic skills such as critical thinking, argumentation and research synthesis (Silva 2015). However, unequal access to support systems across institutions promoted disparities in student engagement and ethical awareness, particularly in under-resourced education settings (Fitzsimons 2024; Naidu and Sevnarayan 2023). Additionally, self-regulation emerged as another critical determinant of ethical GenAI engagement. Students with strong metacognitive skills and a high degree of behavioural control, the main constructs of TPB, were more likely to scrutinise GenAI-generated outputs, maintain academic honesty, and

align their behaviours with high academic expectations (Afzaal et al. 2024; Al-Emran et al. 2025). These findings affirmed the need to integrate self-regulation development into GenAI literacy programs to foster intellectual autonomy and ethical resilience among students.

Moreover, institutional governance was found to significantly affect the effective use of the GenAI tools. The results show that institutions with clear, enforceable GenAI ethics guidelines and transparent communication channels were more successful in handling academic misconduct and promoting responsible innovation (Alexander et al. 2023; R. T. Williams 2023b). However, fragmented policy environments and inconsistent assessment protocols were identified with inequitable academic integrity enforcement and increased misuse of GenAI tools (Currie 2023; Kofinas et al. 2025). Hence, standardising GenAI governance frameworks at national and international levels was recognised as an urgent priority (A. Stone 2023; Tsai 2025; Yang et al. 2024). Furthermore, the review acknowledged several model practices that support responsible GenAI integration in higher education contexts. These included interdisciplinary GenAI ethics committees, discipline-specific faculty professional development initiatives, interactive GenAI literacy modules, and algorithmically transparent GenAI platforms that are co-developed by educators and developers (Perkins, Furze, et al. 2024; Plata et al. 2023). These strategies not only reinforce pedagogical effectiveness but also integrate academic integrity into the teaching and learning processes. Therefore, the ethical and effective use of GenAI in higher education cannot be sustained through isolated interventions. Instead, it requires a coherent ecosystem of academic support, self-regulated student engagement and adaptive institutional governance that is guided by evidence and grounded in inclusive pedagogical values. The proposed integrated TAM-TPB framework provides a useful conceptual scaffold for this endeavour, particularly for institutions in developing regions.

Author Contributions

Daniel Kangwa: conceptualization, data curation, formal analysis, investigation, methodology, project administration, software, supervision, validation, visualization, writing – original draft, writing – review and editing. **Mgambi Msambwa Msafiri:** conceptualization, data curation, formal analysis, investigation, methodology, project administration, resources, software, validation, visualization, writing – original draft, writing – review and editing. **Antony Fute:** data curation, formal analysis, investigation, methodology, project administration, resources, software, validation, visualization, writing – original draft, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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Appendix A

TABLE A1 | Sampled Studies for RQ1.

Academic support	Perceived usefulness	Perceived ease of use	Cultural influence and GenAI effectiveness	Source
Resource Availability	Learning Enhancement	Accessibility	Positive stakeholder attitudes toward GenAI tools can facilitate their integration, but content accuracy remains a concern	(Bobrytska et al. 2024)
			Majority use GenAI tools out of curiosity and necessity; preference for free options	(Chergarova et al. 2023)
Training	Skill Development	User-Friendliness	Positive cultural adaptation to GenAI tools	(Sayed Al Mnhrawi and Alreshidi 2023)
			Identified factors such as knowledge management, trust, and technical resources as critical for GenAI integration	(Razia et al. 2023)
			Reliable and valid measures of GenAI tool usage can help understand their impact on learning in different cultural contexts	(Ahmad et al. 2023)
			Can be engaging and responsive learning tools, but ethical concerns exist	(Chen et al. 2023)
			GenAI-based social learning networks influence perceived usefulness and ease of use	(Saqr et al. 2024)
Technical Guidance	Skill Development	User-Friendliness	Enhance critical thinking and problem-solving skills	(Wang, Lund, et al. 2023)
			Proper training enhances the effectiveness of GenAI tools	(Pham et al. 2023)
			Enhance student engagement and effectiveness in learning	(Barnett-Itzhaki et al. 2023)
			Personalised instructions reduce cognitive load and enhance learning outcomes	(Ferguson et al. 2022)
			Improves feedback analysis	(Sun et al. 2024)
			Improves understanding leads to better engagement and skill development	(Neo 2022)
			Cultural context influences the effectiveness of GenAI tools	(Oluwadiya et al. 2023)
			GenAI tools are versatile and culturally adaptable	(Silva et al. 2024)
			High level of awareness and familiarity	(Yusuf et al. 2024)
			Culturally relevant GenAI tools can motivate students and enhance their learning strategies	(Al-Abdullatif et al. 2023)
Argument maps (AM) can develop argumentative abilities and critical thinking	(Crudele and Raffaghelli 2023)			
Enhance critical thinking and problem-solving skills	(Pacheco-Mendoza et al. 2023)			
Provides lesson plans and support mechanisms, enhancing critical thinking	(van den Berg and du Plessis 2023)			
Affects self-efficacy, creativity, and learning performance	(Wang, Sun, and Chen 2023)			

(Continues)

TABLE A1 | (Continued)

Academic support	Perceived usefulness	Perceived ease of use	Cultural influence and GenAI effectiveness	Source
			Enhance critical thinking by providing interactive and adaptive learning experiences	(Abulibdeh et al. 2024)
			Enhance critical thinking by fostering students' ability to ask questions and analyse information	(Guo and Lee 2023)
			Better development of critical thinking and problem-solving skills	(Opesemowo and Adekomaya 2024)
			Improvement in learning effectiveness and satisfaction levels, especially for low-performing students	(Sayed et al. 2023)
			GenAI-mediated instruction improves English learning outcomes, motivation, and self-regulated learning	(Wei 2023)
			Enhance engagement and support through efficient communication leading to better critical thinking and problem-solving skills	(Antony and Ramnath 2023)
			Higher learning performance and motivation which leads to better critical thinking and problem-solving skills	(Fidan and Gencel 2022)
			Culturally relevant GenAI tools can provide personalised learning experiences, enhancing problem-solving skills	(Ezzaïm et al. 2023)
			Improve learning outcomes in culturally relevant ways	(Shu and Gu 2023)
			Culturally relevant GenAI tools can provide personalised learning experiences, enhancing problem-solving skills	(Alnaqbi and Yassin 2021)
			Ethical challenges for academic integrity, authorship, and publication	(Nam and Bai 2023)
			Positively impacts motivation and academic performance	(Gao et al. 2024)
			GenAI course positively affects students' readiness for medical GenAI	(Taskiran 2023)
			Affects self-efficacy, creativity, and learning performance	(Wang, Lund, et al. 2023)
			Enhance problem-solving skills by providing real-time feedback and personalised learning experiences	(Bosman et al. 2023)
			Accurately respond to chat prompts in problem solving, proving, and explanations	(Remoto 2024)
			Enhanced engagement, improve problem-solving skills	(Essel et al. 2022)
			Autonomous learning with GenAI enhances problem-solving	(Shu and Xu 2022)
			Improves the experience of mobile learners, increasing engagement, understanding, and academic achievements	(Baba et al. 2024)
			GenAI-generated learning materials increase student engagement and study time	(Pesovski et al. 2024)

TABLE A2 | Sampled Studies for RQ2.

Source	Self-regulation	Academic integrity	Behavioural intention to use
(Bouteraa et al. 2024)	Utilisation influenced by performance expectancy, social influence, educational and technology self-efficacy, and personal anxiety	Student integrity negatively impacts usage	Motivation and accountability are key factors
(Hasanein and Sobaih 2023)	Main drivers for using ChatGPT had both positive and negative consequences	Concerns about academic integrity	Mixed intentions based on perceived benefits and risks
(Plata et al. 2023)	Enforcement of academic integrity, education on avoiding misconduct	Encouraging ethical use of GenAI tools	Positive intention to use with proper guidance
(Song 2024)	GenAI tools increase plagiarism and cheating	Need for clear policies and GenAI literacy	Mixed intentions due to ethical concerns
(Zhao et al. 2024)	Significant impact on writing practices	Concerns about plagiarism and technology dependence	High intention to use despite integrity issues
(Chan and Zhou 2023)	Positive correlation between perceived value and intention to use	Weak negative correlation with perceived cost	High motivation to use based on perceived value
(Hualipa et al. 2023)	Recognised value and responsibility of ChatGPT	Stress on privacy and data security standards	Motivation influenced by perceived responsibility
(Rasul et al. 2023)	Benefits like adaptive learning and personalised feedback	Challenges like academic integrity concerns and biases	High intention to use due to adaptive learning benefits
(Sullivan et al. 2023)	Mixed public discussion on GenAI's impact	Focus on academic integrity	Varied intentions based on public perception
(Chai et al. 2024)	Concerns about fairness in educational evaluation	Ethical use emphasised	Motivation influenced by perceived fairness
(Huang 2024)	Improved English pronunciation accuracy	Ethical use in personalised learning paths	High intention to use for language learning
(Alshammari and Alshammari 2024)	Performance expectancy and facilitating conditions influence use	Social influence and effort expectancy less significant	High motivation based on performance expectancy
(Elkhodr et al. 2023)	Positive perception as a learning resource	Improved performance in functionality and comprehension	High intention to use due to perceived benefits
(Rahman et al. 2024)	GenAI-driven summaries improve navigation and usefulness of lecture videos	Ethical use in educational tasks	High intention to use for educational support
(Silva et al. 2024)	Supports various educational tasks	Contributes to sustainability initiatives	High motivation due to sustainability benefits
(Yusuf et al. 2024)	High awareness and familiarity with GenAI tools	Ethical use in sustainable development	High intention to use due to familiarity
(Alamäki et al. 2024)	GenAI literacy in sustainable development	New reference framework for learning tasks	High motivation due to new learning frameworks
(Dai 2024)	Dual-contrast pedagogical approach improves knowledge and skills	Ethical awareness and learning confidence	High intention to use due to improved skills
(Rahiman and Kodikal 2024)	Robust evaluation and assessment methods	Heightened faculty engagement levels	High motivation due to improved assessment methods
(Celik 2023)	Computational thinking as a determinant of GenAI literacy	Ethical use in computational tasks	High intention to use due to computational benefits

(Continues)

TABLE A2 | (Continued)

Source	Self-regulation	Academic integrity	Behavioural intention to use
(Fu et al. 2022)	Improves teaching quality and predicts learning styles	Ethical use in teaching	High motivation due to improved teaching quality
(Boubker 2024)	Output quality influences perceived usefulness	Ethical use in academic writing	High intention to use due to perceived usefulness
(Dergaa et al. 2023)	Enhances academic writing efficiency	Concerns about authenticity and credibility	Mixed intentions due to authenticity concerns
(Saleem et al. 2024)	Applications in personalised learning and assessment	Ethical use in content creation	High intention to use for personalised learning
(Shen et al. 2023)	Immediate scores and diagnostic information aid writing improvement	Ethical use in writing tasks	High motivation due to immediate feedback
(Calonge et al. 2023)	GPT-4 outperforms other chatbots in calculus and statistics	Ethical use in academic evaluation	High intention to use due to performance
(Fan and Li 2023)	Improves quality of art and design education	Ethical use in creative tasks	High motivation due to improved quality
(Rudolph et al. 2023)	GPT-4 performed best among chatbots	Ethical use in educational tasks	High intention to use due to performance
(Sayed et al. 2023)	Improvement in learning effectiveness and satisfaction	Ethical use in learning tasks	High motivation due to improved effectiveness
(Adipat 2023)	Improved interdisciplinary understanding and English-speaking skills	Ethical use in language learning	High intention to use for language skills
(Firat 2023)	Identifies themes like personalised learning and digital literacy	Ethical considerations in digital tasks	High motivation due to personalised learning
(Bobytska et al. 2024)	Positive stakeholder attitudes toward GenAI tools	Concerns about content accuracy	High intention to use despite accuracy concerns
(Chergarova et al. 2023)	Majority use GenAI tools out of curiosity and necessity	Preference for free options	High motivation due to curiosity and necessity
(Sayed AI Mnhrawi and Alreshidi 2023)	Positive cultural adaptation to GenAI tools	Ethical use in cultural contexts	High intention to use due to cultural adaptation
(Razia et al. 2023)	Knowledge management, trust, and technical resources critical for integration	Ethical use in knowledge management	High motivation due to critical integration factors
(Ahmad et al. 2023)	Reliable and valid measures of GenAI tool usage	Ethical use in different cultural contexts	High intention to use due to reliability
(Chen et al. 2023)	Engaging and responsive learning tools	Ethical concerns exist	Mixed intentions due to ethical concerns
(Saqr et al. 2024)	Social learning networks influence perceived usefulness	Ethical use in social learning	High motivation due to perceived usefulness
(Wang, Sun, and Chen 2023)	Enhances critical thinking and problem-solving skills	Ethical use in problem-solving tasks	High intention to use due to skill enhancement
(Pham et al. 2023)	Proper training enhances the effectiveness of GenAI tools	Ethical use in training tasks	High motivation due to training benefits
(Barnett-Itzhaki et al. 2023)	Interactive GenAI tools can enhance student engagement and effectiveness in learning	Ethical use in interactive tasks	High intention to use due to engagement
(Ferguson et al. 2022)	Personalised instructions reduce cognitive load and enhance learning outcomes	Ethical use in personalised learning	High motivation due to reduced cognitive load
(Sun et al. 2024)	AI-BCIFS improves feedback analysis based on EEG signals	Ethical use in feedback tasks	High intention to use due to improved feedback

(Continues)

TABLE A2 | (Continued)

Source	Self-regulation	Academic integrity	Behavioural intention to use
(Neo 2022)	GenAI chatbot improves understanding, leading to better engagement and skill development	Ethical use in understanding tasks	High intention to use due to improved understanding
(Oluwadiya et al. 2023)	Cultural context influences the effectiveness of GenAI tools	Ethical use in cultural contexts	High intention to use due to cultural relevance
(Silva et al. 2024)	GenAI tools are versatile and culturally adaptable	Ethical use in versatile contexts	High intention to use due to adaptability
(Yusuf et al. 2024)	High level of awareness and familiarity with GenAI tools	Ethical use in sustainable development	High intention to use due to familiarity
(Al-Abdullatif et al. 2023)	Culturally relevant GenAI tools can motivate students and enhance their learning strategies	Ethical use in cultural contexts	High intention to use due to cultural relevance
(Crudele and Raffaghelli 2023)	Argument maps (AM) can develop argumentative abilities and critical thinking	Ethical use in argumentative tasks	High motivation due to critical thinking development
(Pacheco-Mendoza et al. 2023)	GenAI tools enhance critical thinking and problem-solving skills	Ethical use in critical thinking tasks	High motivation due to skill enhancement
(van den Berg and du Plessis 2023)	ChatGPT provides lesson plans and support mechanisms, enhancing critical thinking	Ethical use in lesson planning	High intention to use due to support mechanisms
(Abulibdeh et al. 2024)	GenAI tools can significantly enhance critical thinking by providing interactive and adaptive learning experiences	Ethical use in interactive tasks	High intention to use due to interactive learning
(Guo and Lee 2023)	GenAI tools enhance critical thinking by fostering students' ability to ask questions and analyse information	Ethical use in question-asking tasks	High intention to use due to analytical skill enhancement
(Opesemowo and Adekomaya 2024)	GenAI technology leads to better development of critical thinking and problem-solving skills	Ethical use in critical thinking development	High intention to use due to improved problem-solving skills
(Sayed et al. 2023)	Improvement in learning effectiveness and satisfaction levels, especially for low-performing students	Ethical use in effective learning	High motivation due to improved satisfaction
(Wei 2023)	GenAI-mediated instruction improves English learning outcomes, motivation, and self-regulated learning	Ethical use in English learning tasks	High intention to use due to improved outcomes
(Antony and Ramnath 2023)	GenAI chatbots enhance engagement and support through efficient communication, leading to better critical thinking and problem-solving skills	Ethical use in communication tasks	High intention to use due to enhanced support
(Fidan and Gencel 2022)	Chatbot-based and peer feedback groups showed higher learning performance and motivation, which leads to better critical thinking and problem-solving skills	Ethical use in feedback tasks	High intention to use due to performance enhancement
(Ezzaim et al. 2023)	Culturally relevant GenAI tools can provide personalised learning experiences, enhancing problem-solving skills	Ethical use in personalised tasks	High intention to use due to enhanced problem-solving
(Shu and Gu 2023)	GenAI tools improve learning outcomes in culturally relevant ways	Ethical use in culturally relevant learning	High intention to use due to improved outcomes

(Continues)

TABLE A2 | (Continued)

Source	Self-regulation	Academic integrity	Behavioural intention to use
(Alnaqbi and Yassin 2021)	Culturally relevant GenAI tools can provide personalised learning experiences, enhancing problem-solving skills	Ethical use in personalised tasks	High intention to use due to enhanced problem-solving
(Nam and Bai 2023)	Concerns about ethical challenges for academic integrity, authorship, and publication	Ethical use in integrity tasks	High intention to use due to ethical challenges
(Gao et al. 2024)	Technology integration positively impacts motivation and academic performance	Ethical use in performance tasks	High intention to use due to performance improvement
(Taskiran 2023)	GenAI course positively affects students' readiness for medical GenAI	Ethical use in medical GenAI tasks	High intention to use due to readiness improvement
(Wang, Sun, and Chen 2023)	GENAI capability in HEIs significantly affects self-efficacy, creativity, and learning performance	Ethical use in creative tasks	High intention to use due to enhanced creativity
(Bosman et al. 2023)	GenAI tools can enhance problem-solving skills by providing real-time feedback and personalised learning experiences	Ethical use in feedback tasks	High intention to use due to real-time feedback
(Remoto 2024)	GENAI chat models accurately respond to chat prompts in problem-solving, proving, and explanations	Ethical use in problem-solving tasks	High intention to use due to real-time feedback

TABLE A3 | Sampled studies for RQ3.

Source	Institutional policies	Ethical concerns	Example policy to address ethical concerns	Main themes	Self-regulation	Impact on academic integrity	Academic integrity concerns	Reason for behavioural intention to use	Behavioural intention to use
(Abulibdeh et al. 2024)	Technical Guidance	Real-World Connections	Establish interdisciplinary ethics committees	Actual use	Interactive learning experiences	Prevent Integrity biases	Ethical use in feedback tasks	Enhanced learning engagement	High Intention to use
(Adipat 2023)	Technical Guidance	Real-World Connections	Establish interdisciplinary ethics committees	Actual use	Improved interdisciplinary understanding	Improves Ethical Feedback	Ethical use in feedback tasks	Enhanced support	High Intention to use
(Ahmad et al. 2023)	Technical Guidance	Real-World Connections	Establish interdisciplinary ethics committees	Actual use	Reliable and valid measures	Enhanced Ethical Knowledge Management	Ethical use in feedback tasks	Improved skills	High Intention to use
(Al-Abdullatif et al. 2023; Crudele and Raffaghelli 2023; George and Wooden 2023; Habib et al. 2024; Opesemowo and Adekomaya 2024; Silva et al. 2024; Urtasun 2023; Yusuf et al. 2024)	Technical Guidance	Adapt, Support and Engage	Create policies for culturally relevant AI tools	Intention to use	Contextual and cultural relevance motivates and enhance their learning strategies	Prevent Integrity biases	Ethical use in feedback tasks	Despite accuracy concerns	High intention to use
(Alnaqbi and Yassin 2021)	Technical Guidance	Adapt, Support and Engage	Create policies for culturally relevant AI tools	Intention to use	Personalised learning	Enhanced Ethical Awareness	Ethical considerations in digital tasks	Personalised learning	High Intention to use
(Antony and Ramnath 2023)	Technical Guidance	Adapt, Support and Engage	Create policies for culturally relevant AI tools	Intention to use	Efficient communication	Enhanced Ethical Awareness	Ethical use in computational tasks	Perceived usefulness	High Intention to use
(Baba et al. 2024)	Technical Guidance	Adapt, Support and Engage	Create policies for culturally relevant AI tools	Intention to use	Mobile learning	Enhanced Ethical Knowledge Management	Ethical use in lesson planning	Enhanced support	High Intention to use
(Barnett-Itzhaki et al. 2023; Ilieva et al. 2023; Opesemowo and Adekomaya 2024; Pham et al. 2023)	Training	Enhance Contextual Relevance	Develop culturally adaptive GenAI tools to ensure relevance and engagement	Intention to use	Enhance student engagement and effectiveness in learning	Enhanced Ethical Knowledge Management	Ethical use in lesson planning	Enhanced support	High Intention to use

(Continues)

TABLE A3 | (Continued)

Source	Institutional policies	Ethical concerns	Example policy to address ethical concerns	Main themes	Self-regulation	Impact on academic integrity	Academic integrity concerns	Reason for behavioural intention to use	Behavioural intention to use
(Bobrytska et al. 2024; Hu 2023; Silva et al. 2024; Valova et al. 2024)	Training	Enhance Contextual Relevance	Develop adaptive learning frameworks with ethical guidelines	Intention to use	Positive stakeholder attitudes toward GenAI tools	Enhanced Ethical Awareness	Ethical considerations in digital tasks	Personalised learning	High Intention to use
(Bosman et al. 2023)	Training	Enhance Contextual Relevance	Develop integrity policies for AI use	Ethical use	Real-time feedback	Prevent Integrity biases	Concerns about authenticity and credibility	Authenticity concerns	Mixed intentions to use
(Bouteraa et al. 2024)	Training	Concerns about student integrity	Develop integrity policies for AI use	Ethical use	Positive and negative consequences	Enhanced Ethical Awareness	Concerns about authenticity and credibility	Perceived fairness	Mixed intentions to use
(Celik 2023)	Training	Integrate Computational Thinking	Integrate computational ethics into the curriculum	Intention to use	Computational thinking as a determinant of GenAI literacy	Prevent Integrity biases	Concerns about authenticity and credibility	Authenticity concerns	Mixed intentions to use
(Chan and Zhou 2023)	Training	Weak negative correlation between perceived cost and intention to use	Develop cost-benefit analysis policies for AI use	Intention to use	Output quality influences perceived usefulness	Enhanced Ethical Awareness	Concerns about authenticity and credibility	Ethical concerns	Mixed intentions to use
(Chen et al. 2023)	Training	Adapt learning Systems	Establishing ethical guidelines and training programs	Actual use	Engaging and responsive learning tools	Ethical Academic Skills Development	Concerns about authenticity and credibility	Enhanced creativity	High Intention to use
(Chergarova et al. 2023)	Training	Adapt learning Systems	Establishing ethical guidelines and training programs	Actual use	Curiosity and necessity	Improves Ethical Feedback	Concerns about authenticity and credibility	Enhanced support	High Intention to use
(Crudele and Raffaghelli 2023)	Training	Adapt learning Systems	Establishing ethical guidelines and training programs	Actual use	Argument maps	Prevent Integrity biases	Concerns about academic integrity	Perceived benefits and risks	Mixed intentions to use
(Essel et al. 2022)	Training	Adapt learning Systems	Establishing ethical guidelines and training programs	Actual use	Engagement and problem-solving	Prevent Integrity biases	Concerns about academic integrity	Perceived benefits and risks	Mixed intentions to use

(Continues)

TABLE A3 | (Continued)

Source	Institutional policies	Ethical concerns	Example policy to address ethical concerns	Main themes	Self-regulation	Impact on academic integrity	Academic integrity concerns	Reason for behavioural intention to use	Behavioural intention to use
(Ezzaim et al. 2023)	Technical Guidance	Adapt learning Systems	Implement adaptive learning policies with ethical guidelines	Actual use	Adaptive learning and personalised feedback	Prevent Integrity biases	Concerns about plagiarism and technology dependence	Despite integrity issues	High Intention to use
(Ferguson et al. 2022)	Technical Guidance	Tailored inclusive learning	Implement adaptive learning policies with ethical guidelines	Actual use	Personalised instructions	Prevent Integrity biases	Concerns about plagiarism and technology dependence	Despite integrity issues	High Intention to use
(Fidan and Gencel 2022)	Technical Guidance	Tailored inclusive learning	Implement adaptive learning policies with ethical guidelines	Actual use	Learning performance	Enhanced Ethical Awareness	Concerns about plagiarism and technology dependence	Perceived usefulness	High Intention to use
(Gao et al. 2024)	Technical Guidance	Tailored inclusive learning	Implement adaptive learning policies with ethical guidelines	Actual use	Motivation and performance	Concerns about privacy and data security standards	Concerns about plagiarism and technology dependence	Privacy and data security standards	High Intention to use
(Guo and Lee 2023)	Technical Guidance	Tailored inclusive learning	Implement adaptive learning policies with ethical guidelines	Actual use	Questioning and analysis	Academic integrity concerns and biases	Concerns about plagiarism and technology dependence	Adaptive learning benefits	High Intention to use
(Gupta and Chen 2022)	Technical Guidance	Tailored inclusive learning	Develop inclusive AI policies	Intention to use	Personalised learning and digital literacy	Focus on academic integrity	Concerns about plagiarism and technology dependence	Adaptive learning benefits	Mixed intentions to use
(Habib et al. 2024)	Technical Guidance	Balance Support and Autonomy	Create policies balancing AI support and student autonomy	Ethical use	Improves self-efficacy, creativity, and learning performance	Ethical use in academic writing	Concerns about plagiarism and technology dependence	Adaptive learning benefits	High Intention to use
(Hasanein and Sobaih 2023)	Technical Guidance	Concerns about academic integrity	Create academic integrity policies for AI use	Ethical use	Positive and negative consequences	Positive stakeholder attitudes	Concerns about plagiarism and technology dependence	Adaptive learning benefits	High Intention to use
(Hualipa et al. 2023)	Technical Guidance	Concerns about privacy and data security standards	Create privacy and data security policies for AI use	Ethical use	Recognised value and responsibility of ChatGPT	Curiosity and necessity	Ethical considerations in digital tasks	Adaptive learning benefits	High Intention to use
(Liu et al. 2023)	Technical Guidance	Promote Creative Thinking	Establish creative ethics guidelines for AI use	Actual use	Enhances academic writing efficiency	Cultural adaptation	Ethical considerations in digital tasks	Adaptive learning benefits	High Intention to use

(Continues)

TABLE A3 | (Continued)

Source	Institutional policies	Ethical concerns	Example policy to address ethical concerns	Main themes	Self-regulation	Impact on academic integrity	Academic integrity concerns	Reason for behavioural intention to use	Behavioural intention to use
(Luo et al. 2024)	Technical Guidance	Provide Fair Evaluation	Develop fair evaluation policies using AI and blockchain	Ethical use	Concerns about fairness in educational evaluation	Critical factors for integration	Ethical considerations in digital tasks	Adaptive learning benefits	High Intention to use
(Malik, Pratiwi, et al. 2023)	Technical Guidance	Promote Ethical Academic Use	Create policies for ethical use of AI in academic writing	Ethical use	Enhances academic writing efficiency	Understanding impact on learning	Ethical considerations in digital tasks	Adaptive learning benefits	High Intention to use
(Moussa et al. 2024)	Technical Guidance	Encourage Active Learning	Develop active learning policies with ethical AI use	Actual use	Engaging and responsive learning tools	Ethical concerns	Concerns about academic integrity	Adaptive learning benefits	Mixed intentions to use
(Nam and Bai 2023)	Technical Guidance	Academic integrity, authorship, publication	Establishing clear authorship guidelines and integrity policies	Actual use	Ethical challenges	Influence on perceived usefulness and ease of use	Concerns about academic integrity	Perceived ease of use	Mixed intentions to use
(Neo 2022)	Training	Academic integrity, authorship, publication	Establishing clear authorship guidelines and integrity policies	Actual use	Understanding and engagement	Enhance skills	Concerns about academic integrity	Perceived ease of use	Mixed intentions to use
(Oluwadiya et al. 2023)	Training	Academic integrity, authorship, publication	Establishing clear authorship guidelines and integrity policies	Actual use	Cultural context	Enhance effectiveness	Concerns about academic integrity	Perceived ease of use	Mixed intentions to use
(Opesemowo and Adekomaya 2024)	Training	Academic integrity, authorship, publication	Establishing clear authorship guidelines and integrity policies	Actual use	Problem-solving skills	Enhance engagement	Ethical considerations in digital tasks	Perceived ease of use	Mixed intentions to use
(Pacheco-Mendoza et al. 2023)	Training	Encouraging ethical use of GenAI tools	Establishing clear authorship guidelines and integrity policies	Actual use	Critical thinking	Reduce cognitive load	Ethical considerations in digital tasks	Perceived ease of use	Mixed intentions to use
(Pesovski et al. 2024)	Training	Encouraging ethical use of GenAI tools	Establishing clear authorship guidelines and integrity policies	Actual use	Learning materials	Improve feedback analysis	Ethical considerations in digital tasks	Perceived ease of use	High Intention to use
(Pham et al. 2023)	Training	Encouraging ethical use of GenAI tools	Establishing clear authorship guidelines and integrity policies	Ethical use	Proper training	Better engagement and skill development	Ethical considerations in digital tasks	Perceived ease of use	High Intention to use

(Continues)

TABLE A3 | (Continued)

Source	Institutional policies	Ethical concerns	Example policy to address ethical concerns	Main themes	Self-regulation	Impact on academic integrity	Academic integrity concerns	Reason for behavioural intention to use	Behavioural intention to use
(Plata et al. 2023)	Technical Guidance	Encouraging ethical use of GenAI tools	Develop misconduct prevention policies	Ethical use	Enforcement of academic integrity, education on avoiding misconduct	Influence effectiveness	Ethical considerations in digital tasks	Perceived ease of use	High Intention to use
(Rasul et al. 2023)	Technical Guidance	Concerns about biases	Develop bias mitigation policies for AI use	Ethical use	Adaptive learning and personalised feedback	Cultural adaptability	Concerns about content accuracy	Perceived ease of use	High Intention to use
(Razia et al. 2023)	Technical Guidance	Concerns about biases	Develop bias mitigation policies for AI use	Actual use	Knowledge management, trust, technical resources	High awareness	Concerns about content accuracy	Perceived usefulness	High Intention to use
(Remoto 2024)	Technical Guidance	Concerns about biases	Develop bias mitigation policies for AI use	Actual use	Chat prompts	Motivate students	Concerns about content accuracy	Perceived usefulness	High Intention to use
(Saqr et al. 2024)	Technical Guidance	Concerns about biases	Develop bias mitigation policies for AI use	Actual use	Social learning networks	Develop argumentative abilities	Concerns about content accuracy	Perceived usefulness	High Intention to use
(Sayed Al Mnhrwi and Alreshidi 2023)	Technical Guidance	Concerns about biases	Develop bias mitigation policies for AI use	Actual use	Positive cultural adaptation	Enhance skills	Ethical use in lesson planning	Perceived usefulness	High Intention to use
(Shu and Gu 2023)	Technical Guidance	Concerns about biases	Develop bias mitigation policies for AI use	Actual use	Learning outcomes	Enhance critical thinking	Ethical use in lesson planning	Perceived usefulness	High Intention to use
(Shu and Xu 2022)	Technical Guidance	Concerns about biases	Develop bias mitigation policies for AI use	Actual use	Autonomous learning	Foster critical thinking	Ethical use in lesson planning	Perceived usefulness	High Intention to use
(Silva et al. 2024)	Technical Guidance	Creative practical tasks	Create policies for ethical use of AI in coding and content creation	Actual use	Supports various educational tasks	Improve effectiveness	Ethical use in lesson planning	Perceived usefulness	High Intention to use

(Continues)

TABLE A3 | (Continued)

Source	Institutional policies	Ethical concerns	Example policy to address ethical concerns	Main themes	Self-regulation	Impact on academic integrity	Academic integrity concerns	Reason for behavioural intention to use	Behavioural intention to use
(Song 2024)	Technical Guidance	Concerns about plagiarism and technology dependence	Create anti-plagiarism policies and promote AI literacy	Ethical use	Improves writing practices	Improve outcomes	Ethical use in lesson planning	Perceived usefulness	High Intention to use
(Sullivan et al. 2023)	Technical Guidance	Focus on academic integrity	Create anti-plagiarism policies and promote AI literacy	Ethical use	Mixed public discussion on GenAI's impact	Enhance engagement	Ethical use in lesson planning	Perceived usefulness	High Intention to use
(Sun et al. 2024)	Technical Guidance	Focus on academic integrity	Create anti-plagiarism policies and promote AI literacy	Actual use	Improves feedback analysis based on EEG signals	Enhance problem-solving	Ethical use in lesson planning	Perceived usefulness	High Intention to use
(Taskiran 2023)	Training	Practical teaching styles	Develop critical thinking modules with ethical considerations	Actual use	Readiness for medical GenAI	Improve outcomes	Ethical use in lesson planning	Perceived usefulness	High Intention to use
(van den Berg and du Plessis 2023)	Training	Practical teaching styles	Develop critical thinking modules with ethical considerations	Actual use	Provides lesson planning and support mechanisms, enhancing critical thinking	Ethical challenges	Concerns about academic integrity	Perceived usefulness	High Intention to use
(Wang, Sun, and Chen 2023)	Training	Practical teaching styles	Develop critical thinking modules with ethical considerations	Actual use	Critical thinking and problem-solving skills	Positive impact on motivation	Concerns about academic integrity	Perceived usefulness	High Intention to use
(Wei 2023)	Training	Practical teaching styles	Develop critical thinking modules with ethical considerations	Actual use	English learning outcomes	Positive course impact	Concerns about academic integrity	Perceived usefulness	High Intention to use
(Yu and Wang 2024)	Training	Practical teaching styles	Implement training for teachers on ethical AI use	Actual use	Personalised learning and digital literacy	Enhance problem-solving	Ethical use in lesson planning	Perceived ease of use	High Intention to use
(Yusuf et al. 2024)	Training	Concerns about plagiarism and technology dependence	Implement training for teachers on ethical AI use	Actual use	Awareness and familiarity	Accurate responses	Ethical use in lesson planning	Perceived ease of use	High Intention to use
(Zhao et al. 2024)	Training	Concerns about plagiarism and technology dependence	Develop policies to address plagiarism and dependence on AI	Ethical use	Improves writing practices	Enhance engagement	Ethical use in lesson planning	Perceived ease of use	High Intention to use

TABLE A4 | List of the included studies.

S/N	Authors	Focus	Affiliation
1	(Dhamija and Dhamija 2025)	Teachers' perspectives on ChatGPT-generated assignments	India
2	(Ekaterina et al. 2025)	Academic integrity in medical curricula in the AI era	United States
3	(Abdi et al. 2025)	Behavioural intention to use ChatGPT for academics	Italy
4	(Alqahtani and Wafula 2025)	AI integration strategies and policies at universities	United States
5	(Pikhart and Al-Obaydi 2025)	Educators' perspectives on risks of AI in higher education	Iraq
6	(Bayraktar Balkir and Zehir Topkaya 2025)	AI use and academic integrity among EFL learners	Turkey
7	(Alghazo et al. 2025)	ChatGPT's role in higher education and academic integrity	Pakistan
8	(X. Liu 2025)	Detecting AI-generated pseudocode in programming education	United States
9	(Barus et al. 2025)	Student perceptions of AI governance in higher education	Thailand
10	(Farber 2025)	Human versus AI expertise in academic peer review	Israel
11	(Kofinas et al. 2025)	Impact of AI on academic integrity in authentic assessments	United Kingdom
12	(Nguyen et al. 2025)	Ethical and pedagogical principles of GenAI in higher education	Viet Nam
13	(An et al. 2025)	AI guidelines and policies across higher education institutions	United States
14	(Laflamme and Bruneault 2025)	Academic integrity and AI ethics	Canada
15	(Ansone et al. 2025)	GenAI as a transformative tool in art study programs	Latvia
16	(Chan 2025a)	Student perceptions of AI-related academic misconduct	Hong Kong
17	(Alshamy et al. 2025)	Student and academic perceptions of GenAI	Oman
18	(Buniel et al. 2025)	AI dependence and research productivity among STEM students	Philippines
19	(Chan 2025b)	AI integration in nursing education	United States
20	(Velez and Rister 2025)	Institutional anxieties around AI-assisted authorship	United States
21	(Sun, Yang, and Du 2025)	AI-driven total quality management in higher education	China
22	(Hughes et al. 2025)	Challenges of GenAI adoption in higher education	Saudi Arabia
23	(Zlotnikova et al. 2025)	Ethical standards for GenAI in universities	Botswana
24	(Zhao, Cox, and Chen 2025)	AI use by students with disabilities	United Kingdom
25	(Tlais et al. 2025)	AI adoption perspectives among Lebanese STEM faculty	Kuwait
26	(Uppal and Hajian 2025)	Students' perceptions of ChatGPT and its academic effects	Canada
27	(Valdivieso and González 2025)	AI tools in Salvadoran higher education	El Salvador
28	(Ringo 2025)	AI use and doctoral students' research progress	Tanzania
29	(Hasan et al. 2025)	English teachers' perceptions of plagiarism policies	United States
30	(Malik and Terzidis 2025)	Academicians' perspectives on AI in teaching and research	Oman
31	(Wilson 2025)	Development of AI policies in UK universities	Sweden
32	(Evangelista 2025)	Rethinking exam design, assessment strategies, and ethical AI policies in higher education	Russian Federation
33	(Park and Kim 2025)	Ethical AI integration in academic writing	United States
34	(Elizondo-García et al. 2025)	ChatGPT use in mathematics and biology courses with challenge-based learning	Mexico
35	(Nazareth and Choi 2025)	Deterrence and GenAI use in universities	United States
36	(Slimi et al. 2025)	Student perceptions of AI integration in higher education	Spain
37	(Pariyanti et al. 2025)	Balancing AI ethics and spiritual integrity to address cheating	Indonesia
38	(Abdalla 2025)	ChatGPT adoption for data analytics learning among social science students	Oman
39	(Balaskas et al. 2025)	Determinants of ChatGPT adoption in higher education	Greece
40	(Kazley et al. 2025)	Perceptions of ChatGPT use in health professions education	United States

(Continues)

TABLE A4 | (Continued)

S/N	Authors	Focus	Affiliation
41	(Jin et al. 2025)	Institutional adoption policies for GenAI in higher education	Australia
42	(Qu and Wang 2025)	AI guilt and student use of ChatGPT for academic tasks	Singapore
43	(Azoulay et al. 2025)	Ethical challenges and strategic solutions for AI use in CS classrooms	Israel
44	(Waqas et al. 2025)	AIgicism in higher education: AI attitudes and moral disengagement	China
45	(Tamanna and Sinha 2025)	AI in higher education: Opportunities and challenges in Bangladesh	Bangladesh
46	(Yusra and Hanifa 2025)	Academic misconduct in English writing with AI tools	Indonesia
47	(Hickman and Stoica 2025)	Student motivations and attitudes toward ChatGPT use in class	United States
48	(Khodadad 2025)	ChatGPT in engineering education: Opportunities and challenges	Sweden
49	(Altikriti and Nemrawi 2025)	Jordanian students' perceptions of ChatGPT in technical universities	Jordan
50	(Mumtaz et al. 2025)	Ethical use of AI tools in business education	United Arab Emirates
51	(Brummernhenrich et al. 2025)	Social cognition in feedback chatbots	Germany
52	(Al-Jaghoub et al. 2025)	Educators' perceptions of AI in Jordanian higher education	Jordan
53	(Irish et al. 2025)	AI's role in bridging pedagogy and workplace preparation	United States
54	(Stoyanova et al. 2025)	Student and teacher perceptions of ChatGPT in programming education	Bulgaria
55	(Jasrai 2025)	UTAUT model analysis of ChatGPT use among Indian students	India
56	(Cordero and Cordero-Castillo 2025)	AI integration in rural Philippine higher education	Philippines
57	(Evangelista 2025)	Academic integrity and assessment redesign in the age of ChatGPT	United Arab Emirates
58	(Mulaudzi and Hamilton 2025)	Lecturer perspectives on AI and personalised learning	South Africa
59	(Acosta-Enriquez et al. 2025)	AI use by researchers: Social influence, self-efficacy, and integrity	Peru
60	(Asad and Aijaz 2025)	Impact of ChatGPT and GenAI on skill enhancement in higher education	Pakistan
61	(Buitrago-Ciro et al. 2025)	AI integration and outreach in academic libraries	Canada
62	(Nogueira et al. 2025)	Evaluation of AI-driven academic assessment tools	Norway
63	(Guruge et al. 2025)	Framework to address and prevent contract cheating	Australia
64	(Symeou et al. 2025)	Evidence-based guidelines for GenAI integration in universities	Cyprus
65	(Adiyono et al. 2025)	AI's impact on student reliance for exam answers	Indonesia
66	(Almaghaslah 2025)	AI-generated MCQs in pharmacy education	Saudi Arabia
67	(Corbin et al. 2025)	Framework for acceptable AI uses in assessment	Australia
68	(Hossain et al. 2025)	AI literacy among LIS students in Bangladesh, India, and Pakistan	Pakistan
69	(Fošner 2024)	University students' attitudes toward AI tools for education	Slovenia
70	(Johnson et al. 2024)	Anticipated impact of AI on higher education	United States
71	(Kutty et al. 2024)	Perspectives of students, educators, and administrators on GenAI	Sri Lanka
72	(Acosta-Enriquez et al. 2024)	Generation Z university students' knowledge, attitudes, and ethics on ChatGPT	Peru
73	(McIntire et al. 2024)	Ethics of academic integrity and plagiarism	United States
74	(Yusuf et al. 2024)	AI and higher education: Threat versus reformation	Spain
75	(Rudolph et al. 2024)	Higher education's AI paradox and chatbot adoption	Australia
76	(Elbaz et al. 2024)	Business student perceptions of morality and AI in Oman	Oman
77	(Shabbir et al. 2024)	AI's educational potential in underdeveloped regions	Malaysia
78	(Bouteraa et al. 2024)	AI diffusion in higher education and student integrity concerns	Malaysia
79	(Teremetskyi et al. 2024)	Academic integrity and AI: Ukraine's legal perspective	Ukraine

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S/N	Authors	Focus	Affiliation
80	(Katsamakakos et al. 2024)	AI-driven transformation of higher education institutions	United States
81	(Driessens and Pischetola 2024)	Danish university policies on GenAI	Denmark
82	(Bikanga Ada 2024)	Computer science students' perceptions of ChatGPT for learning	United Kingdom
83	(Streletska et al. 2024)	AI integration into STEM education and academic integrity	Ukraine
84	(Liu, Park, and McMinn 2024)	Student perspectives on AI/ChatGPT in academic communication	Hong Kong
85	(Song 2024)	Higher education crisis: AI-related academic misconduct	United States
86	(Perkins, Roe, Vu, et al. 2024)	Bypassing GenAI text detectors: Implications for inclusive education	Viet Nam
87	(Rasul et al. 2023)	Framework to enhance academic integrity in the GenAI era	Malaysia
88	(Luo et al. 2024)	AI and blockchain in scoring and data integrity	China
89	(Bannister et al. 2023)	Generative AI in transnational education: Policy development	Spain
90	(Tang and Eaton 2024)	AI-generated content footprints in scholarly publications	Canada
91	(Bhullar et al. 2024)	ChatGPT in higher education: Literature synthesis and research agenda	Australia
92	(Almassaad et al. 2024)	Student perceptions of generative AI in higher education	Saudi Arabia
93	(Xu, Su, et al. 2024)	Learners' perceptions of ChatGPT in China	China
94	(Slimi and Carballido 2023)	Expert perspectives on AI integration in higher education	Spain
95	(Klyshbekova and Abbott 2024)	ChatGPT in assessment: A disruptor or enhancement?	United Kingdom
96	(Zhou et al. 2024)	Student experiences with AI in higher education	United Kingdom
97	(Veras et al. 2024)	AI usability in health sciences education	Canada
98	(Katsantonis and Katsantonis 2024)	University students' AI attitudes: Cognitive, emotional, and behavioural analysis	United Kingdom
99	(Hsiao and Tang 2024)	Determinants of GenAI-supported learning in higher education	Taiwan
100	(Asad and Ajaz 2024)	AI's impact on lifelong learning and upskilling in higher education	Pakistan
101	(Mwakapina 2024)	AI's role in language teaching and learning in higher education	Tanzania
102	(Castelló-Sirvent et al. 2024)	Roadmap for AI and ethics in universities	Portugal
103	(Khlaif et al. 2025)	University teachers' views on AI integration for student assessment	Palestine
104	(Revell et al. 2024)	AI versus human essayists: Impact on authorship and academic integrity	United Kingdom
105	(Scarfe et al. 2024)	AI infiltration in university examination systems: A case study	United Kingdom
106	(Olla et al. 2024)	Enhancing human-AI interaction in large language models	United States
107	(Kumar et al. 2024)	AI advancements in higher education: A tri-phased exploration	India
108	(Lee et al. 2024)	Educator perspectives on GenAI in higher education	Australia
109	(Smerdon 2024)	AI in essay-based assessment: Adoption and student performance	Australia
110	(Nguyen and Goto 2024)	Unmasking AI-related academic cheating in Vietnamese universities	Japan
111	(Suleymenova et al. 2024)	AI in economics education: Challenges and ethical dilemmas	United Kingdom
112	(Rapi et al. 2024)	AI adoption in Islamic higher education	Indonesia
113	(Malik, Hassan, et al. 2023)	ChatGPT in medical education: RAG-assisted anticoagulation management	United States
114	(Farooqui et al. 2024)	Framing assessment questions with AI-generated content	India
115	(Huo and Siau 2024)	AI in business higher education: Focus group study	Singapore
116	(Perkins, Roe, Postma, et al. 2024)	Detecting GPT-4 generated text: AI misuse in higher education	Viet Nam
117	(Nikolic et al. 2025)	AI integration framework for project-based learning	Australia
118	(Lund et al. 2025)	AI and academic integrity: Student perceptions	United States
119	(Akiba and Garte 2024)	Leveraging AI in university writing instruction	United States

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S/N	Authors	Focus	Affiliation
120	(Niloy et al. 2024)	AI chatbots and academic integrity concerns	Bangladesh
121	(Krecar et al. 2024)	Perceptions of ChatGPT for homework: Students' and professors' perspectives	Croatia
122	(Gruenhagen et al. 2024)	Academic integrity concerns and students' use of AI chatbots for assessments	Australia
123	(Wu et al. 2024)	AI governance case studies in Big Ten universities	United States
124	B. W. Stone (2024)	Generative AI in higher education: Uncertainty and perspectives	United States
125	(Sykes 2024)	AI-assisted academic writing: Comparative study of student versus AI-enhanced critiques	Canada
126	(P. Williams 2023a)	Comparing GenAI performance on undergraduate and postgraduate assessments	United Kingdom
127	(Susnjak et al. 2022)	ChatGPT and online exam integrity	Australia
128	(Amrane-Cooper et al. 2024)	AI in online assessment: Ethical considerations	United Kingdom
129	(Isiaku et al. 2024)	Academic evolution in the age of ChatGPT: Ethics and learning impact	Nigeria
130	(Ullah et al. 2020)	Guidelines for AI tools in the world's top 50 universities	Portugal
131	(Wang, Huai, et al. 2024)	AI governance in higher education: Policy analysis	United States
132	(Waltzer et al. 2024)	Identifying AI-generated writing in college essays	United States
133	(Asamoah et al. 2024)	DEQ framework for generative AI in education and knowledge work	Ghana
134	(Espinoza Vidaurre et al. 2024)	AI and academic integrity perceptions among students in Peru and Chile	Peru
135	(Awadallah Alkhouk and Khlaif 2024)	AI-resistant assessments: Faculty training insights	Palestine
136	(Murray et al. 2015)	Postsecondary instructor perceptions of AI-generated papers	Canada
137	(Wang, Huai, et al. 2024)	AI-generated content's impact on college students' learning ability	China
138	(Cooper et al. 2024)	Large language models for coding, teaching, and inclusion in research	United States
139	(Vetter et al. 2024)	AI ethics framework: Text generators and academic integrity	United States
140	(Bukar et al. 2024)	AI decision-making framework in education: Case study on ChatGPT	Iraq
141	(Perkins 2023)	Academic integrity considerations for AI language models	Viet Nam
142	(Barrientos et al. 2024)	Discourse analysis on AI academic integrity: Science and mathematics student perspectives	Philippines
143	(Farrokhnia et al. 2024)	SWOT analysis of ChatGPT: Implications for education	Netherlands
144	(Sullivan et al. 2023)	ChatGPT in higher education: Academic integrity and learning	Australia
145	(Bannister et al. 2023)	Disrupting generative AI discourse with marginalised epistemic capital	United Kingdom
146	(Întorsureanu et al. 2024)	AI and academic integrity in economics dissertations	Romania
147	(Cotton et al. 2024)	AI-driven cheating concerns and academic integrity in education	United Kingdom
148	Nikolic et al. (2023)	ChatGPT versus engineering education assessments	Australia
149	(Khan et al. 2024)	AI trustworthiness in higher education	Finland
150	(Cavazos et al. 2024)	Student perspectives on AI classroom integration	United States
151	(Nartey 2024)	AI guiding principles for employability in UK universities	United Kingdom
152	(Pham et al. 2023)	AI-assisted digital transformation in engineering education	Viet Nam
153	(Butson and Spronken-Smith 2024)	AI's impact on higher education research	New Zealand
154	(Gregory and Narang 2024)	AI for learning: Generative AI experimentation at the University of Miami	United States
155	(Yang and Stivers 2024)	AI language models solving undergraduate finance problems	United States

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S/N	Authors	Focus	Affiliation
156	(Muthukrishnan et al. 2024)	Prospective teachers' AI literacy and responsible AI use	Iraq
157	(Williams and Fadda 2023)	Evaluating ChatGPT's performance in glycobiology assessments	Ireland
158	(Alharbi et al. 2024)	Bridging teacher awareness and student AI adoption gaps in language education	Saudi Arabia
159	(Hammond et al. 2023)	Critical discourse analysis of AI paraphrasing sites	New Zealand
160	(Wise et al. 2024)	Scholarly dialogue: Writing, authorship, and AI integrity	Australia
161	(Hargreaves 2023)	ChatGPT's impact on law school assessments	Hong Kong
162	(Raman et al. 2024)	University students' adoption of ChatGPT via innovation theory	India
163	(Cowling et al. 2023)	AI and authorship policies for academic publishing	Australia
164	(Rienties et al. 2025)	AI digital assistant preferences among distance-learning students	United Kingdom
165	(A. Stone 2023)	Student perceptions of academic integrity: Understanding consequences	United Kingdom
166	(Currie 2023)	A conversational engagement with ChatGPT	Australia
167	(Naidu and Sevnarayan 2023)	AI and online assessment in distance education	South Africa
168	(Tyler et al. 2024)	AI in criminal justice education: Educator perspectives	United States
169	(Mugambiwa 2024)	Evaluating AI's impact on academic writing in higher education	South Africa
170	(Malik, Hassan, et al. 2023)	AI's role in student-written essays: Higher education perspective	Indonesia
171	(Alexander et al. 2023)	AI-generated writing detection in second-language higher education	Cyprus
172	(Currie and Barry 2023)	AI use in nuclear medicine education	Australia
173	(Soos and Haroutunian 2024)	AI authorship debates and ethical implications	United States
174	(Ghimire et al. 2024)	ChatGPT's implications for higher education: Nepali student perspectives	Nepal
175	(Plecerda 2024)	Academic integrity surrounding generative AI: ICT student lenses	Philippines
176	(Furze et al. 2024)	AI Assessment Scale (AIAS): Pilot implementation in GenAI-supported assessments	Australia
177	(Boudia and Krismadinata 2024)	SLA with ChatGPT: Computer science teachers and students' perspectives	Indonesia
178	(Mirta et al. 2024)	ChatGPT's impact on research and knowledge advancement	Paraguay
179	(Nam and Bai 2023)	Ethical implications of ChatGPT in STEM research and higher education	China
180	(Rasul et al. 2023)	Role of ChatGPT in higher education: Benefits, challenges, and future directions	Australia
181	(Cowling et al. 2023)	AI replacing humans in higher education: Student retention and success	Australia
182	(Saqib and Zia 2024)	Evaluating AI-generated content for academic integrity verification	Pakistan
183	(Slotnick and Boeing 2024)	Enhancing qualitative research in higher education via AI integration	United States
184	(Fu et al. 2024)	Assessing ChatGPT's information quality using user satisfaction models	Taiwan
185	(Ardito et al. 2020)	Generative AI detection in higher education assessments	United Kingdom
186	(Waluyo and Kusumastuti 2024)	AI-enhanced English learning outcomes in Thai higher education	Thailand
187	(Elhersh and Alqawasmeh 2024)	Public sentiment and ethical concerns regarding ChatGPT in universities	Jordan
188	(Whitfield and Hofmann 2023)	Elicit: AI-assisted literature review research tool	United States
189	(Tarisayi 2024)	ChatGPT adoption in South African universities: A socio-technical analysis	South Africa
190	(Jafry and Vorstermans 2024)	AI, disability, and academic integrity in higher education	Canada

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S/N	Authors	Focus	Affiliation
191	(Crawford et al. 2023)	Ethical leadership and ChatGPT in assessment and learning	Australia
192	(P. Williams 2023a)	Ethical considerations of using generative chatbots in higher education	United Kingdom
193	(Laker and Sena 2023)	Accuracy and detection of student ChatGPT usage in business analytics courses	Philippines
194	(Chaudhry et al. 2023)	Reevaluating student performance assessment in the ChatGPT era	United Arab Emirates
195	(Villarreal-Torres et al. 2023)	Generative AI's challenges and opportunities in higher education	United Kingdom
196	(Shanto et al. 2023)	PAIGE: AI-based framework for enhancing assignment integrity	Bangladesh
197	(Shrivastava 2024)	Transformative role of ChatGPT in academic discourse	India
198	(Margono et al. 2024)	Virtual tutoring with ChatGPT and its impact on Indonesian academia	Malaysia
199	(Gonsalves 2024)	Addressing student non-compliance in AI declarations	United Kingdom
200	(Bukar et al. 2024)	Integrative framework for regulating ChatGPT usage	Nigeria
201	(Matthews and Volpe 2023)	Academics' perceptions of ChatGPT-generated text in Turing's Imitation Game	Australia
202	(Moorhouse et al. 2023)	AI tools and assessment guidelines in top-ranking universities	China
203	(ElFangary 2024)	Roadmap for AI-driven learning in Egyptian higher education	Egypt
204	(Kostka and Toncelli 2023)	ChatGPT applications in English language teaching	United States
205	(Fidas et al. 2023)	AI-centred proctoring systems and academic integrity in online learning	Greece
206	(Bannister et al. 2023)	English Medium Instruction and AI-based assessment literacy	Spain
207	(Karkoulia et al. 2024)	Academic integrity perceptions of ChatGPT in higher education	Lebanon
208	(Azevedo et al. 2014)	AI policies, equity, and faculty morality implications in higher education	United States
209	(Currie and Barry 2023)	ChatGPT in medical imaging education	Australia
210	(Heugh et al. 2022)	Translation technology for multilingual education in EMI universities	Australia
211	(Verhoef et al. 2022)	Enhancing academic integrity via communities of practice	South Africa
212	(Henry and Oliver 2022)	Ethical dimensions of AI-assisted academic proctoring	United Kingdom
213	(Köbis and Mehner 2021)	Ethical concerns of AI-supported mentoring in higher education	Germany